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Assignment Title: Investigating Neighbourhood
Unemployment in England Through Machine Learning
and Spatial Analysis

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Abstract

Unemployment remains an important socio-economic challenge in England, with substantial variation between neighbourhoods associated with deprivation, labour-market participation, transport accessibility, housing and demographic characteristics. This study investigates the socio-economic characteristics associated with unemployment across English Lower Layer Super Output Areas (LSOAs) using machine learning and spatial analysis.

Official Census 2021, English Indices of Multiple Deprivation (IMD) 2019 and population-density data were integrated to produce an analytical dataset of 31,810 English LSOAs with 16 final predictors. Linear Regression, Random Forest, XGBoost and LightGBM were compared using RMSE, MAE, R^2 and Adjusted R^2 . Feature importance and SHAP were used for model interpretation, while QGIS examined geographical patterns of unemployment and selected predictors.

The ensemble models consistently outperformed Linear Regression. Tuned XGBoost achieved the strongest performance, with RMSE of 1.1033, MAE of 0.8391, R^2 of 0.8207 and Adjusted R^2 of 0.8202, although differences between the ensemble models were relatively small. Never Worked Rate, Income Deprivation Score and No Car or Van Rate emerged as the most influential predictors, while migration indicators provided predictive information but were less influential than the leading structural characteristics. QGIS analysis further demonstrated substantial geographical variation and identified neighbourhoods where high unemployment coincided with high labour-market disengagement, income deprivation and lack of car or van access.

Overall, the findings indicate that neighbourhood unemployment in England is multidimensional, predictively nonlinear and geographically uneven. The study demonstrates the value of combining explainable machine learning with spatial visualisation while interpreting neighbourhood-level relationships as associations and predictive contributions rather than causal effects.

Keywords: Unemployment; Machine Learning; XGBoost; SHAP; Spatial Analysis; QGIS; LSOA; Socio-economic Determinants; England.

List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
GIS	Geographic Information System
IMD	Index of Multiple Deprivation
LGBM	Light Gradient Boosting Machine
LSOA	Lower Layer Super Output Area
MAE	Mean Absolute Error
ML	Machine Learning
ONS	Office for National Statistics
QGIS	Geographic Information System (GIS) software
R²	Coefficient of Determination
RMSE	Root Mean Squared Error
SHAP	SHapley Additive exPlanations
SQL	Structured Query Language
UK	United Kingdom
VIF	Variance Inflation Factor
XGBoost	Extreme Gradient Boosting

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1. CHAPTER 1: INTRODUCTION

1.1. Research Background

Unemployment remains a significant socio-economic challenge in England, closely linked to household income, labour-market participation and inequalities between places. However, unemployment is unevenly distributed across the country. Neighbourhoods vary substantially in deprivation, educational attainment, health, housing, transport accessibility, demographic composition and other socio-economic characteristics, resulting in marked geographical variation in labour-market outcomes.

Much existing research examines unemployment at national, regional or local-authority level. While these scales are useful for understanding broad labour-market patterns, they may obscure important differences between smaller communities. Lower Layer Super Output Areas (LSOAs) offer a more detailed neighbourhood-level geography, enabling analysis of unemployment and associated socio-economic conditions across small and consistently defined areas. This is important because neighbourhoods within the same local authority can experience very different socio-economic circumstances and labour-market outcomes.

1.2. Research Problem

Previous research has identified a range of characteristics associated with unemployment, such as deprivation, education, health, housing, transport accessibility, migration and demographic composition. However, these factors are often examined in isolation or at broader geographical scales. There is therefore a need to analyse multiple socio-economic characteristics together at neighbourhood level within a consistent analytical framework.

This study integrates official Census 2021, English Indices of Multiple Deprivation (IMD) 2019 and population data at LSOA level to address this gap. Machine learning is applied to investigate predictive relationships and compare modelling approaches, while QGIS is used to analyse geographical patterns. As the analysis is based on observational area-level data, findings are interpreted as associations and predictive contributions rather than causal effects.

1.3. Research Aim

This study aims to investigate the socio-economic determinants of unemployment across England using neighbourhood-level data, machine learning and spatial analysis.

1.4. Research Objectives

The objectives are:

1. To construct an integrated England-wide LSOA dataset using official Census, deprivation and population data.
2. To examine relationships between unemployment and key socio-economic characteristics, including migration indicators.
3. To develop and compare predictive models of neighbourhood unemployment using a consistent evaluation framework.

4. To identify the variables making the greatest contribution to unemployment prediction.
5. To examine geographical patterns of unemployment and selected socio-economic characteristics using QGIS.

1.5. Research Questions

RQ1. Which socio-economic characteristics are most strongly associated with unemployment across English LSOAs?

RQ2. What predictive contribution do migration indicators make when considered alongside other socio-economic characteristics?

RQ3. Which modelling approach provides the strongest predictive performance for unemployment across English LSOAs?

RQ4. Which variables have the greatest influence on predictions produced by the best-performing model?

RQ5. What geographical patterns of unemployment and its key socio-economic correlates can be identified using QGIS?

1.6. Significance of the Study

This study presents an England-wide, neighbourhood-level analysis of unemployment, integrating multiple socio-economic dimensions within a consistent data science framework. Employing LSOAs allows for greater geographical resolution than regional approaches, while machine learning facilitates the investigation of complex predictive relationships. Model interpretation and spatial visualisation are used to identify which characteristics are most influential and to examine where unemployment and related disadvantages are concentrated. The findings contribute evidence on neighbourhood labour-market inequality, with appropriate caution that area-level associations reflect predictive relationships rather than individual or causal effects.

2. CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

Unemployment is unevenly distributed across the UK, and research increasingly treats this as a spatial as well as an economic issue. Lee, Fransham and Bukowski (2024) emphasise the role of labour-market institutions and structural change, whereas Overman and Xu (2024) place greater weight on the spatial concentration of skills and employment opportunities (Lee, Fransham and Bukowski, 2024; Overman and Xu, 2024). Although both perspectives establish that place matters, neither implies that broad regional patterns fully explain variation within smaller neighbourhoods.

The literature therefore needs to move from national and regional inequality towards neighbourhood conditions. Existing studies have examined deprivation, education, health, housing, accessibility, migration and demographic characteristics, but differences in populations, geographical units and methods make direct comparison challenging. Machine learning enables the evaluation of multiple predictors at once, although most unemployment research has focused on national forecasting or individual risk rather than current English neighbourhoods. This review first considers spatial labour-market inequality, then neighbourhood-level determinants, and finally comparative machine learning, before outlining limitations and identifying the specific research gap at the LSOA level.

2.2. Unemployment and Spatial Labour-Market Inequality

Gilmartin and Korobilis (2012), using UK regional panel data and a mixture modelling approach, identified different patterns of high and low unemployment and showed that factors such as sectoral structure and urbanisation were associated differently across regions (Gilmartin and Korobilis, 2012). Their findings are important because they demonstrate that unemployment does not follow a uniform geographical pattern. However, the study was conducted at a relatively broad regional scale, which may conceal substantial differences between smaller neighbourhoods. More recent research provides complementary explanations for these spatial inequalities. Lee, Fransham and Bukowski (2024) emphasise the role of economic restructuring and labour-market institutions, whereas Overman and Xu (2024), using British local labour-market data and decomposition analysis, highlight spatial differences in skills and the concentration of highly skilled workers. Although these studies differ in their methods and explanations, collectively they indicate that geographical inequalities in unemployment reflect multiple interconnected factors rather than a single mechanism.

Smaller-scale studies reveal additional complexity. Fieldhouse and Tranmer (2001), using 1991 British Census microdata and multilevel modelling, found that much neighbourhood variation in male unemployment was associated with housing-market selection rather than purely contextual neighbourhood effects (Fieldhouse and Tranmer, 2001). The multilevel approach is a strength because it distinguishes individual from area processes. However, the cross-sectional and historical dataset limits contemporary and causal interpretation.

Vandecasteele and Fasang (2021), using UK Household Longitudinal Study data linked to neighbourhood employment deprivation, found that deprivation prolonged unemployment primarily where individuals' friendship networks were concentrated locally (Vandecasteele and Fasang, 2021). In contrast to a uniform neighbourhood effect, their results imply that place

interacts with social context. However, the outcome is individual unemployment exit rather than an area unemployment rate. Plum and Knies (2019) similarly show, through dynamic random-effects panel modelling, that local unemployment modifies transitions from low pay, but their contextual measure operates at local-authority rather than LSOA level (Plum and Knies, 2019).

Lloyd, Norman and McLennan (2023) examined long-term small-area deprivation in England and demonstrated substantial persistence in geographical disadvantage. However, constructing comparable measures across several decades requires the harmonisation of changing indicators and geographical units (Lloyd, Norman and McLennan, 2023). This methodological challenge illustrates why contemporary LSOA-level analysis is valuable for examining current socio-economic patterns.

Collectively, the spatial literature establishes that disadvantage is geographically persistent, but it does not determine which combinations of contemporary neighbourhood characteristics provide the strongest predictive information about unemployment. This shifts the review from where unemployment inequality occurs to which neighbourhood characteristics are associated with it.

2.3. Neighbourhood-Level Socio-economic Determinants

Deprivation is an important candidate, but its measurement is not straightforward. Radburn and Beecham (2021) show that small-area visualisation can identify deprivation pockets obscured by broader geographical representations (Radburn and Beecham, 2021). However, their approach is descriptive and does not determine which deprivation dimensions are most relevant to labour-market outcomes.

Watson et al. (2019), comparing expert-derived IMD weights with alternative preference-based approaches, found broad agreement but also evidence that some weights could reasonably differ (Watson *et al.*, 2019). This weakens the assumption that a composite deprivation score is necessarily the most informative representation for every research outcome. Knies and Kumari (2022), using more than 25,000 Understanding Society respondents, found that multimorbidity associations were concentrated particularly within income, education, employment and health deprivation. Their dependent variable is health rather than unemployment, so the result cannot be transferred directly (Knies and Kumari, 2022). Nevertheless, both studies indicate that deprivation domains contain distinct information and support their separate evaluation.

Skills and health also appear interconnected. Overman and Xu (2024) identify skills concentration as an important element of labour-market inequality, whereas Banks, Karjalainen and Waters (2024) show that working-age disability differs substantially by education, cohort and region (Banks, Karjalainen and Waters, 2024). Murray et al. (2016), analysing 98,756 ONS Longitudinal Study members, found that poor health and higher local unemployment were independently associated with subsequent workforce exit (Murray *et al.*, 2016). The longitudinal design provides stronger temporal evidence than a cross-sectional comparison, but the geography is local-authority level, and the dependent variable is workforce exit rather than neighbourhood unemployment.

Ayoubkhani et al. (2025) provide stronger population coverage by linking Census, NHS, HMRC and DWP records and applying quasi-experimental methods to health-employment

relationships (Ayoubkhani *et al.*, 2025). However, their estimates still depend on matching and identification assumptions. Banks, Karjalainen and Waters (2024), by contrast, use descriptive comparisons and therefore provide weaker evidence about causal pathways. Together, the studies support inclusion of health and disability characteristics while cautioning against treating them as direct causes of area unemployment.

Transport accessibility provides a more explicitly spatial mechanism. Bastiaanssen, Johnson and Lucas (2022) combined British employment microdata with nationwide public-transport accessibility measures and used instrumental-variable methods to address endogeneity. They found higher accessibility was associated with employment particularly among lower-income, lower-educated and low-car-ownership groups (Bastiaanssen, Johnson and Lucas, 2022). This provides stronger evidence than simple correlation, but the study remains cross-sectional and its outcome is individual employment probability. Its relevance is therefore to justify testing accessibility-related variables, not to assume their importance at LSOA level.

Housing is similarly difficult to isolate because Fieldhouse and Tranmer (2001) demonstrate that tenure and residential sorting can contribute to apparent neighbourhood effects (Fieldhouse and Tranmer, 2001). A housing or transport variable that predicts unemployment strongly may consequently represent overlapping socioeconomic processes rather than an independent mechanism.

Clark *et al.* (2019), using Quarterly Labour Force Survey data linked with local-area deprivation measures, examined the labour-market integration of new migrants to England and identified variation in employment outcomes across migrant groups and local contexts (Clark *et al.*, 2019). However, the analysis used a broader geographical context than LSOA level, which may conceal differences between smaller neighbourhoods. This limitation supports examining migration alongside other socio-economic characteristics at a finer geographical scale rather than interpreting migrant status as an independent explanation of unemployment. Jivraj and Alao (2023), linking Understanding Society with IMD and Census information, found persistent employment penalties for some ethnic groups but little evidence that deprivation or ethnic density systematically removed these disadvantages (Jivraj and Alao, 2023).

Rather than establishing migration or ethnicity as isolated explanations, these studies show that outcomes depend on individual and contextual characteristics. Their implication for this research is therefore comparative: migration and ethnicity should be evaluated alongside deprivation, education, health, housing and accessibility rather than assigned a direction or importance beforehand.

2.4. Machine Learning and Comparative Modelling

The multidimensional nature of neighbourhood unemployment creates a modelling challenge because relationships may be nonlinear and predictors may interact. Linear Regression remains useful because it provides a transparent benchmark for a continuous target. Its coefficients allow comparatively direct interpretation, while its additive structure tests how far a simpler functional form can explain neighbourhood variation. Its limitation is that nonlinearities and interactions must generally be specified explicitly. It is therefore an appropriate baseline rather than a deliberately weaker comparator.

Random Forest offers a different strategy. Gogas, Papadimitriou and Sofianos (2022), using 36 euro-area macroeconomic indicators, found Random Forest achieved the strongest out-of-sample directional forecasting performance among the methods tested (Gogas, Papadimitriou and Sofianos, 2022). However, their task involved aggregate time-series classification rather than cross-sectional LSOA regression. Celbiş (2023), using rural European individual-level data, combined tree-based machine learning with SHAP and identified heterogeneous relationships across demographic and educational groups (Celbiş, 2023). Yet the rural European and individual-level setting also limits direct generalisation to English neighbourhoods. These studies justify testing Random Forest, not assuming it will be superior.

Boosting provides a second nonlinear comparison. Zhao (2020) combined ensemble prediction of long-term unemployment with SHAP interpretation and reported strong predictive performance, but also identified fairness concerns affecting some population groups (Zhao, 2020). Tu and Tran (2025), using UK monthly unemployment observations, found XGBoost stronger than ARIMA, ARIMAX and Random Forest for their forecasting task (Tu and Tran, 2025). However, national temporal forecasting differs fundamentally from neighbourhood-level regression. Their findings support evaluating XGBoost, not favouring it before testing.

LightGBM provides an alternative boosting architecture. Ke et al. (2017) developed LightGBM to improve the efficiency and scalability of gradient-boosted decision trees, while He, Zhu and Fu (2022) applied LightGBM to predicting employment outcomes among college graduates (Ke Guolin *et al.*, 2017; He, Zhu and Fu, 2022). However, graduate employability differs substantially from neighbourhood-level unemployment in both the unit of analysis and prediction problem. Therefore, this evidence supports the methodological suitability of LightGBM for structured predictive modelling, rather than providing direct evidence that it will perform strongly for LSOA-level unemployment.

The comparison of Linear Regression, Random Forest, XGBoost and LightGBM is consequently purposeful. Linear Regression provides an interpretable linear benchmark; Random Forest evaluates a bagging-based nonlinear ensemble; XGBoost and LightGBM evaluate alternative boosting strategies. Comparing them using the same observations, predictors and metrics is more informative than comparing performance values reported across unrelated datasets.

Viljanen and Pahikkala (2020) further demonstrate that machine learning can improve labour-market risk prediction using administrative records (Viljanen and Pahikkala, 2020). However, the study operates at the individual level, reinforcing the need to test whether similar predictive advantages occur with area-level data. Interpretability remains necessary. Lundberg and Lee (2017) developed SHAP as a theoretically grounded framework for feature attribution. Its value is especially relevant when flexible tree ensembles outperform more transparent statistical models (Scott M. Lundberg; Su-In Lee, 2017). However, SHAP explains the fitted model rather than the causal data-generating process. Feature importance should therefore be interpreted as predictive contribution, not evidence that changing a neighbourhood characteristic would necessarily change unemployment.

2.5. Limitations of Existing Approaches and Research Gap

The reviewed literature highlights three main limitations. First, geographical scale differs substantially between studies. Gilmartin and Korobilis (2012) and Overman and Xu (2024) identify broad spatial inequalities in labour-market outcomes, but regional or local labour-

market analysis may obscure variation between smaller neighbourhoods. In contrast, Radburn and Beecham (2021) demonstrate that small-area analysis can reveal local deprivation patterns that are less apparent at broader scales, although their approach remains primarily descriptive. This suggests that findings from larger geographical units cannot be assumed to apply at LSOA level and supports analysis at a finer spatial scale.

Second, evidence is fragmented across socio-economic dimensions. Murray et al. (2016) examine health and workforce exit, Bastiaanssen, Johnson and Lucas (2022) investigate transport accessibility and employment, and Clark et al. (2019) focus on migrant labour-market integration. While these studies provide important evidence within their respective domains, differences in datasets, outcomes, geographical units and methodologies limit direct comparison. Consequently, the literature does not clearly establish the relative predictive contribution of multiple socio-economic characteristics when evaluated together within the same neighbourhood-level dataset.

Third, predictive and spatial research often address different analytical questions. Machine-learning studies such as Gogas, Papadimitriou and Sofianos (2022) and Tu and Tran (2025) demonstrate the potential of nonlinear models for unemployment analysis, but their applications focus on broader forecasting rather than cross-sectional prediction across English neighbourhoods. Small-area studies offer greater spatial detail but typically do not compare alternative predictive models under consistent conditions. Existing evidence therefore supports the use of machine learning but does not establish which modelling approach is most suitable for LSOA-level unemployment.

The research gap is not a lack of studies on unemployment, socio-economic disadvantage, machine learning or spatial inequality. Instead, relatively few studies have combined England-wide LSOA-level analysis, multiple socio-economic domains, a common modelling dataset, comparative Linear Regression, Random Forest, XGBoost and LightGBM modelling, predictive interpretation and spatial visualisation within a single analytical framework. This integration enables consistent comparison of socio-economic characteristics and examination of how unemployment and key predictors vary geographically.

The present study addresses this narrower gap by integrating Census 2021, IMD 2019 and population-density data at LSOA level. The four regression models are compared using consistent evaluation measures, feature importance and SHAP are used to interpret predictive contributions, and QGIS is used to examine spatial patterns. The contribution lies in the integration of fine-scale geography, multidimensional socio-economic analysis, comparative predictive modelling, model interpretation and spatial visualisation, rather than in any individual method alone.

2.6. Chapter Summary

Recent literature refines the research focus from broad spatial labour-market inequality to neighbourhood-level variation. Studies examining deprivation, skills, health, housing, accessibility, migration and ethnicity suggest that labour-market outcomes are associated with multiple interconnected socio-economic characteristics. However, differences in geographical scale, study populations, outcome definitions and methodologies limit direct comparison between findings and prevent a clear assessment of the relative importance of each characteristic.

The review demonstrates the potential of machine learning for analysing complex labour-market relationships, while highlighting the trade-off between predictive flexibility and interpretability. This research gap provides a rationale for an England-wide LSOA-level framework that consistently evaluates multiple socio-economic characteristics, compares Linear Regression, Random Forest, XGBoost and LightGBM, interprets their predictive contributions and examines relevant geographical patterns. Chapter 3 applies this synthesis to data acquisition, data preparation, variable selection, comparative modelling, model evaluation and spatial analysis.

Author(s)	Year	Country	Study Focus	Data / Study Area	Method / Model	Main Findings	Limitations	Relevance to This Study
Gilmartin and Korobilis	2012	UK	Spatial unemployment	UK regions	Mixture model	Unemployment patterns vary geographically.	Broad regional scale.	Supports finer-scale analysis.
Lee, Fransham and Bukowski	2024	UK	Spatial inequality	UK labour markets	Quantitative analysis	Structural factors contribute to spatial inequality.	Broad geographical focus.	Establishes spatial context.
Overman and Xu	2024	UK	Spatial inequality	Local labour markets	Decomposition	Skills contribute to geographical inequality.	Above LSOA scale.	Supports skills variables.
Fieldhouse and Tranmer	2001	Britain	Housing and unemployment	Census microdata	Multilevel model	Residential selection contributes to unemployment differences.	Historical cross-sectional data.	Supports housing context.
Vandecasteele and Fasang	2021	UK	Neighbourhood disadvantage	UKHLS	Longitudinal analysis	Local networks modify deprivation effects.	Individual-level outcome.	Supports neighbourhood analysis.
Lloyd, Norman and McLennan	2023	England	Deprivation	Small areas	Longitudinal analysis	Spatial disadvantage persists over time.	Changing measures/geographies.	Supports small-area deprivation analysis.
Radburn and Beecham	2021	England	Small-area deprivation	English small areas	Spatial visualisation	Fine-scale mapping reveals local deprivation patterns.	Mainly descriptive.	Supports spatial visualisation.
Knies and Kumari	2022	UK	Deprivation domains	UKHLS	Statistical modelling	Deprivation domains contain distinct information.	Health outcome.	Supports domain-level variables.
Murray et al.	2016	UK	Health and employment	ONS Longitudinal Study	Longitudinal analysis	Health and local unemployment relate to workforce exit.	Not LSOA unemployment.	Supports health variables.
Bastiaanssen, Johnson and Lucas	2022	Britain	Transport accessibility	National microdata	IV analysis	Accessibility is associated with employment.	Individual employment outcome.	Supports accessibility variables.

Clark et al.	2019	England	Migration and employment	QLFS/local deprivation	Statistical analysis	Migrant outcomes vary by group and context.	Broader geography.	Supports migration variables.
Jivraj and Alao	2023	UK	Ethnicity and employment	UKHLS/IMD/Census	Statistical modelling	Employment inequalities persist across some ethnic groups.	Individual-level analysis.	Supports ethnicity variables.
Gogas et al.	2022	Euro area	Unemployment prediction	Macroeconomic data	RF and other ML	RF performed strongly in prediction.	Aggregate forecasting task.	Supports testing RF.
Celbiş	2023	Europe	Labour-market prediction	Rural Europe	Tree ML + SHAP	ML captures heterogeneous relationships.	Different population/context.	Supports nonlinear ML.
Zhao	2020	Europe	Unemployment prediction	Individual data	Ensemble ML + SHAP	ML supports prediction and interpretation.	Individual-level analysis.	Supports explainable ML.
Tu and Tran	2025	UK	Unemployment forecasting	UK monthly data	ARIMA/RF/XGBoost	XGBoost performed strongly.	National time-series task.	Supports testing XGBoost.
Ke et al.	2017	General	LightGBM	Benchmark datasets	LightGBM	Efficient gradient-boosted tree learning.	Not unemployment-specific.	Justifies LightGBM.
He, Zhu and Fu	2022	Employment	Employment prediction	Graduate data	LightGBM	LightGBM applied to employment prediction.	Different outcome/population.	Supports testing LightGBM.
Viljanen and Pahikkala	2020	Labour market	Unemployment risk	Administrative data	Machine learning	ML can predict unemployment risk.	Individual-level prediction.	Supports comparative ML.
Lundberg and Lee	2017	General	Model interpretation	ML applications	SHAP	SHAP explains feature contributions.	Does not establish causality.	Supports SHAP interpretation.

Table 2.1: Literature Synthesis Matrix

3. CHAPTER 3: METHODOLOGY

3.1. Introduction

This chapter outlines the methodological framework used to investigate the socio-economic characteristics associated with unemployment across English neighbourhoods. The approach is aligned with the research aim, objectives and research questions set out in Chapter 1, and is informed by the limitations identified in the Literature Review. Evidence from previous research indicates that unemployment varies geographically, that neighbourhood labour-market outcomes are associated with multiple interconnected socio-economic characteristics, and that predictive and spatial approaches are often considered separately.

A quantitative secondary-data design will be used to integrate neighbourhood-level socio-economic, deprivation and geographical information within a single analytical framework. The methodology includes data acquisition and integration, data quality assessment, feature engineering, exploratory analysis, variable selection, comparative regression modelling, model interpretation and spatial analysis. The analysis aims to identify which neighbourhood characteristics are associated with unemployment, compare predictive approaches, assess the contribution of migration alongside wider socio-economic conditions, and examine relevant geographical patterns. As the study uses observational area-level data, the analysis is designed to identify associations and predictive contributions rather than causal effects.

3.2. Research Design

This research adopts a quantitative, observational and predictive design. A quantitative approach is appropriate as unemployment and the explanatory characteristics can be represented using numerical rates, percentages and deprivation scores. This allows relationships to be examined systematically across a large number of neighbourhoods and enables competing predictive models to be evaluated using common performance criteria.

Secondary data are appropriate as nationally standardised neighbourhood datasets are available from official UK sources. These data provide much broader geographical coverage than could be achieved through primary surveys and allow neighbourhoods across England to be assessed within a consistent framework. The analysis is conducted at Lower Layer Super Output Area (LSOA) level, as broader regional or local-authority statistics may conceal differences between smaller communities, which was identified as a recurring limitation in Chapter 2. The analysis components correspond directly to the research questions. Statistical association and model interpretation will support RQ1, while retaining migration variables alongside deprivation, education, health, housing, transport and demographic indicators will enable RQ2 to assess migration in context rather than independently. Comparison of Linear Regression, Random Forest, XGBoost and LightGBM will address RQ3. QGIS-based analysis will address RQ5 by examining geographical patterns, while feature importance and SHAP will support RQ4 by assessing which predictors contribute most strongly to the best-performing model. These questions are explicitly distinguished in Chapter 1.

3.3. Study Area, Data Sources and Acquisition

The study focuses on England at LSOA level. Multiple official datasets will be combined because unemployment is unlikely to be adequately represented through a single socio-

economic domain. The source data collectively cover labour-market status, education, health, disability, migration, ethnicity, housing, accessibility, occupational structure, industry, deprivation and geographical context.

Dataset	Provider	Year	Coverage	Expected source observations	Main variables	Target	Suitability
Economic Activity	ONS Census	2021	England & Wales LSOAs	35,672	Unemployment and inactivity indicators	Unemployment_Rate	Provides the study outcome and labour-market measures
Age Structure	ONS Census	2021	England & Wales LSOAs	35,672	Age groups	No	Supports demographic indicators
Highest Qualification	ONS Census	2021	England & Wales LSOAs	35,672	Qualification levels	No	Represents skills and human capital
General Health & Disability	ONS Census	2021	England & Wales LSOAs	35,672	Health and disability	No	Represents health-related labour-market context
Migration & Ethnicity	ONS Census	2021	England & Wales LSOAs	35,672	Migration origin and ethnic groups	No	Supports RQ2 and population composition analysis
Housing & Accommodation	ONS Census	2021	England & Wales LSOAs	35,672	Tenure and accommodation	No	Represents housing conditions
Car Availability & Travel to Work	ONS Census	2021	England & Wales LSOAs	35,672	Car access and commuting	No	Represents accessibility and mobility
NS-SeC & Industry	ONS Census	2021	England & Wales LSOAs	35,672	Occupational and industrial structure	No	Represents socio-economic and employment structure
Population Density	ONS	2021	LSOAs	35,672	Population per km ²	No	Provides settlement context
Rural-Urban Classification	ONS	2021	LSOAs	35,672	Rural/urban classification	No	Provides geographical context

English IMD	MHCLG	2019	England LSOAs	England small areas	Deprivation and domain scores	No	Captures multidimensional neighbourhood disadvantage
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Table 3.1: Summary of Data Sources Used in the Study

The expected final analytical dataset will contain approximately 31,810 matched English LSOAs. The target variable will be `Unemployment_Rate`, while the initial candidate variables will represent the socio-economic dimensions identified in Chapter 2.

Census 2021 is suitable because it provides detailed and nationally standardised neighbourhood information across the majority of the required socio-economic domains. IMD 2019 complements the Census because it provides established measures of neighbourhood deprivation, including income, education, health, crime, housing and living-environment dimensions that cannot be represented fully through Census indicators alone. Population density and rural-urban classification provide additional geographical context.

3.3.1. Data Acquisition and Integration

Datasets are sourced from official ONS and UK Government data services and downloaded in tabular format. Census 2021 provides labour-market and socio-economic characteristics, while IMD 2019, population-density and geographical classification data add deprivation and spatial context. All sources are prepared at a common LSOA geographical level to ensure consistent integration.

The datasets are integrated using `LSOA_Code` as the common geographical identifier. Economic Activity serves as the base labour-market dataset as it contains the target `Unemployment_Rate`. Demographic, qualification, health, disability, migration, ethnicity, transport, housing, occupational, industrial, population-density, region/rural-urban and deprivation variables are matched using their corresponding LSOA codes.

A relational left-join strategy is applied, retaining the Economic Activity dataset as the reference and appending variables where LSOA codes match. The analysis is restricted to English LSOAs. This approach integrates information from multiple sources while maintaining a consistent neighbourhood unit. Matching success, unmatched records and duplicate identifiers are assessed during data-quality validation.

3.4. Data Preparation and Feature Engineering

Data preparation is required because variables collected from different datasets are not automatically comparable or suitable for predictive modelling. The aim is therefore to create consistent measures while retaining genuine socio-economic variation.

Feature engineering will transform source variables into measures that are comparable across LSOAs and meaningful for unemployment analysis. This is necessary because many Census variables originate as population or household counts. Direct comparison of counts could reflect differences in neighbourhood population size rather than socio-economic composition. Relevant variables will therefore be converted into rates or percentages using appropriate denominators, enabling more meaningful comparison between differently sized LSOAs.

Derived measures will also be constructed where grouped indicators better represent concepts identified in the Literature Review. These will include young, working-age and elderly population rates, qualification indicators and migration measures distinguishing movement within the UK from migration originating outside the UK. Such transformations allow demographic, human-capital and migration characteristics to be evaluated alongside other neighbourhood conditions within a consistent modelling framework.

Geographical context will also be considered because labour-market relationships may vary spatially (Gilmartin and Korobilis, 2012). Region and rural-urban information obtained from the geographical lookup data are categorical rather than continuous variables. Where required for analysis, these categories will therefore be converted into appropriate numerical representations rather than treating administrative codes as meaningful quantities. Overlapping geographical classifications will be reviewed to avoid introducing redundant information.

3.5. Data Quality, Exploratory Analysis and Data Treatment

Before modelling, the data will be checked for missing observations, duplicated LSOA codes, inconsistent data types and implausible values. These checks are essential to ensure that errors arising from data acquisition or integration are not misinterpreted as genuine neighbourhood variation.

Exploratory Data Analysis will then assess central tendency, variability, skewness, outliers and the relationships between unemployment and candidate predictors. Summary statistics, histograms, density plots, boxplots and predictor-target plots will be used as they provide complementary insights into the data structure.

Extreme observations require particular attention in socio-economic data, as high unemployment or deprivation may reflect genuine neighbourhood disadvantage rather than data errors. Extreme values will not be removed automatically. Where necessary, Winsorisation will be applied to reduce undue influence while retaining neighbourhood observations. Predictors that remain substantially positively skewed will be considered for logarithmic transformation. This approach is intended to improve analytical robustness without excluding neighbourhoods that represent meaningful social inequalities.

3.6. Variable Selection and Multicollinearity

Variable selection will combine evidence from Chapter 2, correlation analysis, redundancy assessment and Variance Inflation Factor (VIF) analysis.

Correlation analysis will give an initial indication of relationships between candidate predictors and unemployment, and will help identify variables that carry similar information. However, predictors will not be selected solely on the basis of their bivariate correlations. This is especially relevant for migration, as RQ2 requires its predictive contribution to be assessed alongside broader structural conditions, regardless of whether migration has the strongest simple association with unemployment.

Variables representing the same underlying concept will be reviewed prior to modelling. The analytical framework already identifies overlapping IMD measures, car-availability indicators and complementary disability measures as potential sources of redundancy.

VIF will then be used to assess multicollinearity among numerical candidates. This is important for Linear Regression, where strong linear relationships among explanatory variables can destabilise coefficient estimates and reduce interpretability. VIF will be used as a diagnostic tool rather than an automatic exclusion rule, with statistical evidence considered alongside theoretical relevance.

3.7. Predictive Modelling and Validation

The final predictor dataset will be separated into training and testing samples so that model development and model evaluation remain distinct. The same partition will be used for all four models to provide a fair comparative framework.

Four regression approaches are selected because they represent different modelling assumptions.

Linear Regression models a continuous response as a linear combination of explanatory variables and provides an interpretable baseline. Its purpose is to determine how well a relatively simple linear model performs before introducing more flexible approaches.

Random Forest is an ensemble method that combines predictions from multiple decision trees constructed using randomisation and bootstrap sampling (Breiman, 2001). It is suitable because neighbourhood unemployment may involve nonlinear relationships and interactions between socio-economic characteristics.

XGBoost is a gradient-boosting approach in which trees are developed sequentially so that later learners improve errors remaining from previous learners (Chen and Guestrin, 2016). It is included to evaluate whether sequential boosting improves prediction relative to Linear Regression and Random Forest.

LightGBM is a gradient-boosted decision-tree framework developed for efficient learning on structured datasets (Ke Guolin *et al.*, 2017). Comparing LightGBM with XGBoost allows two boosting approaches to be evaluated under the same conditions rather than assuming that one boosting implementation will automatically be superior.

The four-model comparison therefore represents a deliberate progression:

linear benchmark → bagging ensemble → gradient boosting → alternative gradient boosting.

The ensemble models will additionally undergo hyperparameter tuning using cross-validation within the training data. This allows parameter choices to be evaluated without repeatedly consulting the independent test set and reduces the risk of selecting configurations that generalise poorly.

Standardisation will be applied where methodologically required for Linear Regression, while tree-based models will not require equivalent feature scaling.

3.8. Model Evaluation and Interpretation

Predictive performance will be compared using RMSE, MAE, R^2 and Adjusted R^2 . RMSE gives greater weight to larger prediction errors, MAE represents average absolute prediction error, R^2 describes the proportion of variation represented by the model, and Adjusted R^2 provides an additional measure accounting for the predictor set.

Multiple evaluation metrics are used as each reflects a distinct aspect of predictive performance, providing a more comprehensive assessment than reliance on a single measure.

Feature importance will subsequently be used to examine the relative contribution of predictors within the tree-based models. SHAP analysis will provide a more detailed interpretation of predictor contributions to model outputs. This is directly relevant to RQ4 and to the interpretability concern identified in Chapter 2. Neither feature importance nor SHAP will be interpreted as evidence of causality.

3.9. Spatial Analysis, Reliability, Ethics and Limitations

QGIS will be used to address RQ5. While machine-learning models provide information about predictive relationships, they do not directly indicate the locations of high and low unemployment. The analytical dataset will be linked to official LSOA boundaries using LSOA_Code.

Choropleth maps will present the geographical distribution of unemployment and selected socio-economic variables. Bivariate maps will examine the geographical co-occurrence of unemployment with selected influential characteristics. Machine learning identifies predictive importance, while QGIS identifies geographical patterns, providing complementary analytical perspectives.

Reliability will be supported by the use of official data sources, documented integration procedures, geographical validation, consistent model partitions and evaluation measures. Ethical risk is limited, as only publicly available aggregated neighbourhood data will be analysed rather than identifiable individual records. Variables such as ethnicity, migration and disability will be interpreted cautiously to avoid attributing area-level associations to individual residents.

Methodological limitations include the observational and cross-sectional design, the temporal difference between Census 2021 and IMD 2019, potential differences in geographical matching between datasets, and the possibility that some relevant neighbourhood factors are not captured in the available data.

3.10. Research Workflow and Chapter Summary

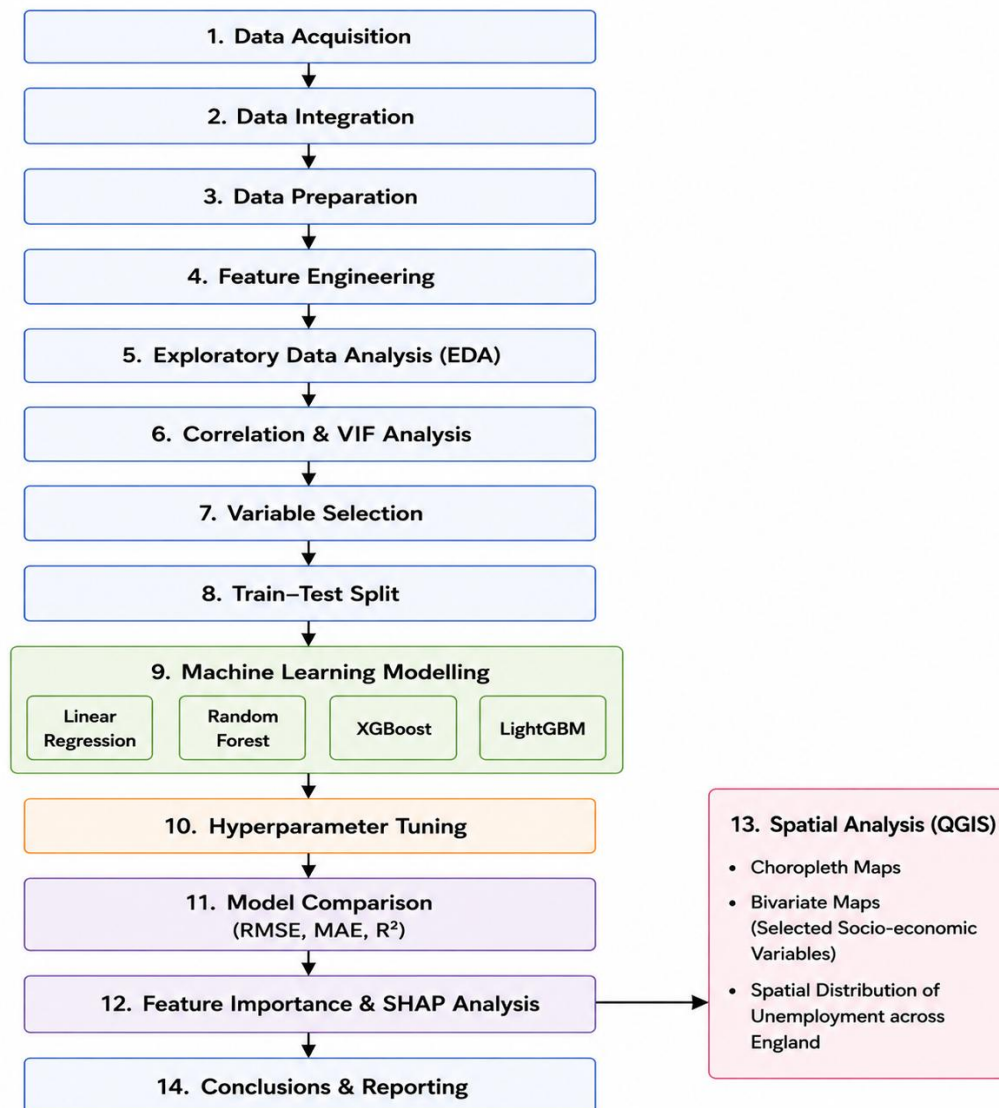


Figure 3.1: Overall Research Workflow of the Study

Source: Author's own illustration.

The above figure summarises the overall research workflow, progressing from data acquisition and preparation to feature engineering, variable selection, machine-learning modelling, model evaluation and interpretation, followed by QGIS spatial analysis.

This chapter has established and justified the methodological framework for investigating neighbourhood-level unemployment across England. The approach integrates official socio-economic and spatial data, feature engineering, exploratory analysis, variable selection, comparative regression modelling, model interpretation and spatial analysis. Together, these methods provide a consistent framework for addressing the research questions. Chapter 4 presents the practical implementation of this methodology using SQL, Python and QGIS, while Chapter 5 evaluates and critically discusses the resulting findings.

4. CHAPTER 4: IMPLEMENTATION

4.1. Introduction

This chapter details the practical implementation of the analytical framework established in Chapter 3. Building on the methodology, it describes the application of the selected datasets, preprocessing procedures, machine-learning models and spatial methods, and presents the main outputs from the analysis.

The implementation followed a structured sequence: data integration, raw-data inspection, feature engineering, data treatment, exploratory data analysis (EDA), variable selection, machine-learning development, model evaluation, model interpretation and spatial visualisation. DB Browser for SQLite supported data integration, Python in Google Colab was used for data preparation and machine learning, and QGIS was used for spatial analysis. Instead of including large blocks of source code, tables and figures are provided as evidence of the implemented work. The complete supporting code and project files are available via the links in the Appendix.

The final analytical dataset contained 31,810 English Lower Layer Super Output Areas (LSOAs). Four regression algorithms were implemented: Linear Regression, Random Forest, XGBoost and LightGBM. Their predictive performance was compared using common evaluation metrics, followed by feature-importance and SHAP analysis. Finally, QGIS was used to examine the geographical distribution of unemployment and selected socio-economic predictors.

4.2. Data Integration and Preparation

4.2.1. Software Environment

The analytical workflow required different software environments for different stages. DB Browser for SQLite was used to integrate the original datasets through their common geographical identifier. Google Colab provided the Python environment for preprocessing, EDA, variable selection, modelling and model interpretation. QGIS was used to connect the analytical results with England LSOA boundaries and produce spatial outputs.

Software/Library	Implementation purpose
DB Browser for SQLite	Dataset integration
Google Colab	Python analytical environment
Pandas and NumPy	Data preparation and feature engineering
Matplotlib and Seaborn	EDA and visualisation
Scikit-learn	Preprocessing, Linear Regression and Random Forest
XGBoost	Gradient-boosting regression
LightGBM	Gradient-boosting regression
SHAP	Model interpretation

QGIS	Spatial analysis and mapping
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Table 4.1: Software and Tools Used

This structure enabled the analysis to progress consistently from raw official datasets to statistical, predictive and geographical outputs.

4.2.2. Data Integration and Raw Inspection

The implementation began by integrating the Census 2021, IMD 2019, population-density and geographical reference data described in Chapter 3. Each of the 15 Census 2021 source datasets contained 35,672 LSOA records for England and Wales. As this study focuses on England, the data were matched with England-specific sources and restricted to the relevant geography.

LSOA_Code was used as the common geographical identifier during SQL integration. The Economic Activity dataset provided the base labour-market structure. Demographic, qualification, health, disability, migration, ethnicity, transport, housing, occupation, industry and other socio-economic variables were appended by matching LSOA codes. IMD 2019 and population-density measures were then incorporated at the same geographical level.

After geographical matching and restriction to England, the final integrated dataset contained 31,810 LSOAs. The reduction from 35,672 source LSOAs to 31,810 analytical observations primarily reflects the England-only study coverage and matching of the required geographical records, rather than deletion of observations during later modelling.

Initial inspection was undertaken in Python. Dataset dimensions, data types, missing values, duplicate records, duplicate LSOA identifiers and numerical ranges were examined. No missing values, duplicate rows or duplicated LSOA codes were identified in the final integrated dataset.

Implementation stage	Observations/Result
Census 2021 source LSOAs, England and Wales	35,672
Final matched England LSOAs	31,810
Difference following geographical restriction/matching	3,862
Missing values in final dataset	0
Duplicate rows	0
Duplicate LSOA codes	0

Table 4.2: Dataset Integration and Quality Summary

These checks confirmed that the integrated dataset was structurally complete before feature engineering and analytical preprocessing.

4.2.3. Feature Engineering and Data Treatment

Feature engineering converted Census counts into comparable rates and percentages. This prevented differences in neighbourhood population size from dominating comparisons between LSOAs. Derived measures represented labour-market status, age structure, qualifications, migration, disability, ethnicity, housing and transport. Categorical geographical information

was also converted into an analytical representation where required, while identifiers and redundant classifications were excluded from numerical modelling.

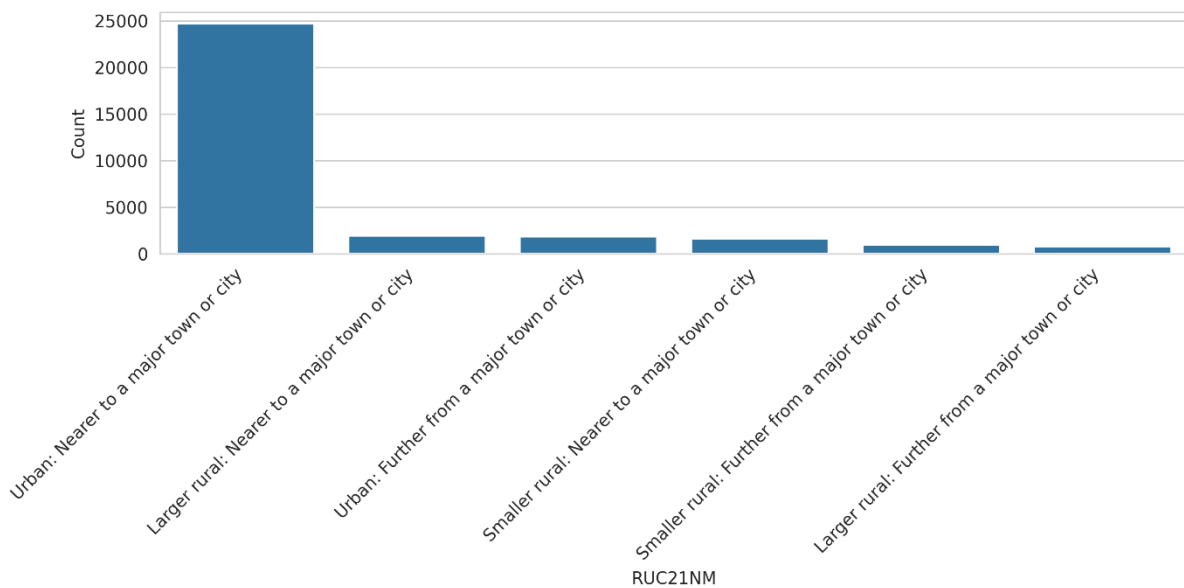


Figure 4.1: Distribution of LSOAs by Detailed Rural-Urban Classification

The above figure shows the distribution of LSOAs across the six detailed rural-urban categories. Most LSOAs are classified as urban areas nearer to a major town or city, while substantially fewer fall within rural or more remote urban categories. This demonstrates the predominantly urban composition of the study area and provides geographical context for the analysis.

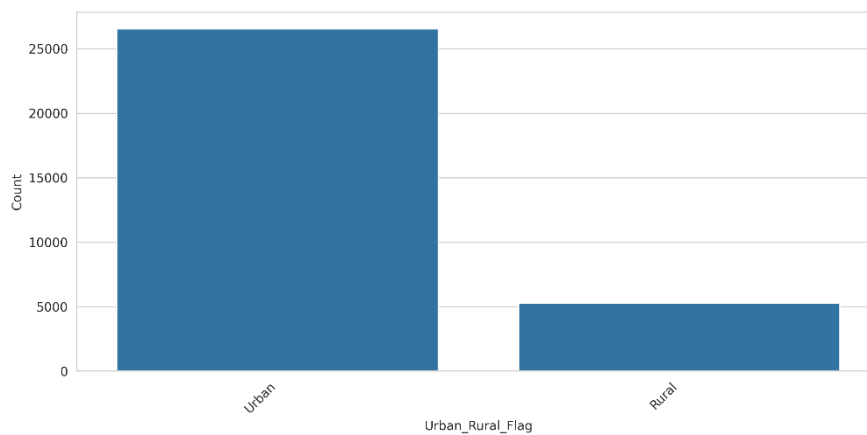


Figure 4.2: Distribution of LSOAs by Simplified Urban-Rural Classification

The above figure shows the distribution of LSOAs according to the simplified Urban-Rural Flag. The dataset is predominantly urban, comprising 26,531 urban LSOAs and 5,279 rural LSOAs. This highlights the strong urban representation within the study dataset and provides geographical context for the subsequent unemployment analysis.

Following feature engineering, numerical variables were assessed for extreme observations and skewness. One-per cent winsorisation was applied to control extreme values while retaining the corresponding LSOAs. Predictors remaining substantially positively skewed after winsorisation were transformed using log1p where appropriate. The target unemployment measure was also treated consistently according to the preprocessing framework established in Chapter 3.



Figure 4.3: Distribution of Unemployment Rate Before and After Data Treatment

The above Figure demonstrates the effect of treatment on the dependent variable. The overall distribution was preserved while the influence of extreme observations was controlled.

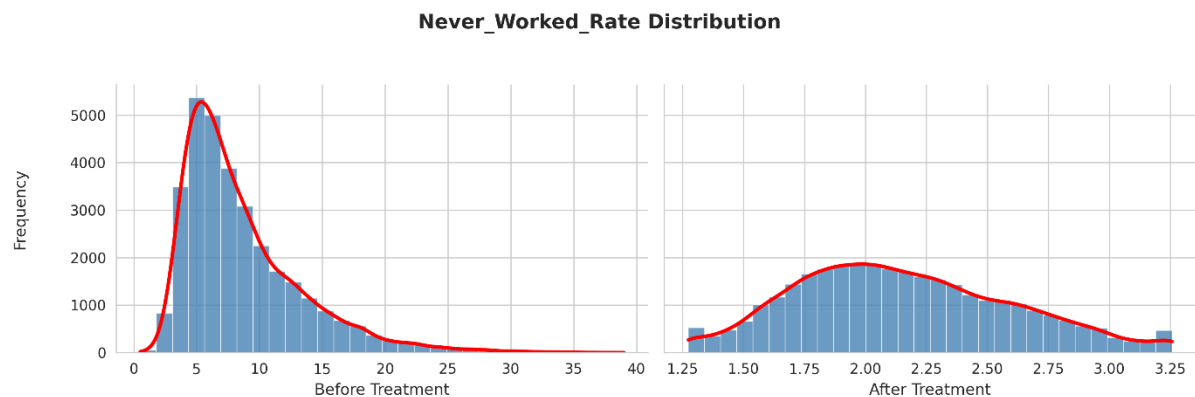


Figure 4.4: Distribution of Never Worked Rate Before and After Data Treatment

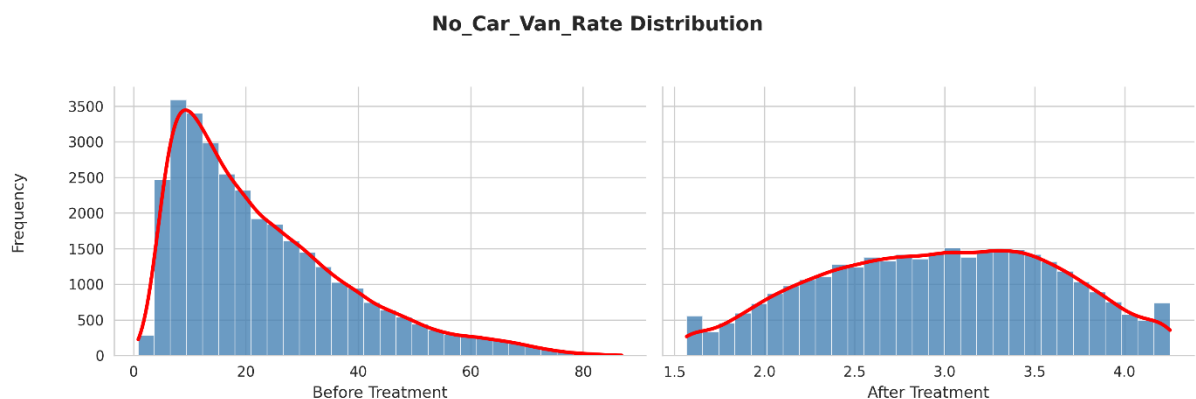


Figure 4.5: Distribution of No Car Van Rate Before and After Data Treatment

The before-and-after comparison demonstrates that preprocessing reduced extreme skewness in selected predictors without removing the neighbourhood observations themselves. This provides visual evidence of the implemented data-treatment procedure.

4.3. Exploratory Data Analysis and Variable Selection

4.3.1. Exploratory Data Analysis

EDA was implemented to examine variable distributions, identify extreme observations and assess initial relationships between unemployment and the candidate predictors. Histograms, density plots, boxplots and predictor-versus-target scatterplots were generated. As the complete analysis produced a large number of figures, representative outputs are presented in this section, while the complete analytical workflow is available through the code link provided in the Appendix.

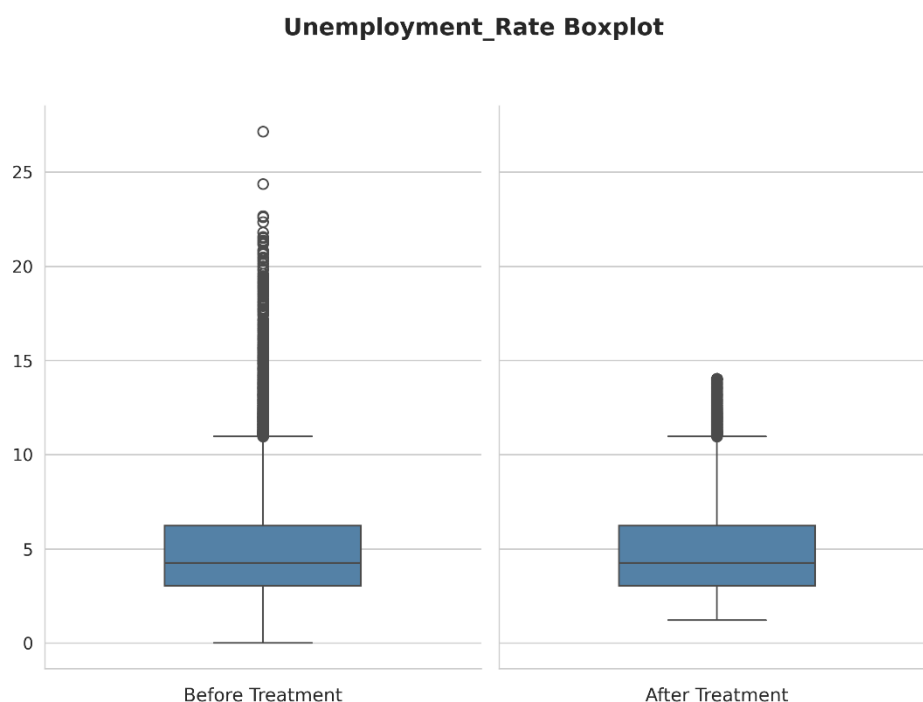


Figure 4.6: Unemployment Rate Boxplot Before and After Data Treatment

The boxplot illustrates the presence of extreme unemployment values before treatment and the reduced influence of these observations following preprocessing. The comparison also confirms that the central distribution of unemployment was retained rather than removing LSOAs with relatively high values.



Figure 4.7: Relationship Between Unemployment Rate and Never Worked Rate Before and After Data Treatment

A clear positive relationship remained visible after treatment, indicating that LSOAs with higher never-worked rates generally also experienced higher unemployment. The persistence of this relationship after preprocessing provided initial evidence of the relevance of this variable for subsequent modelling.

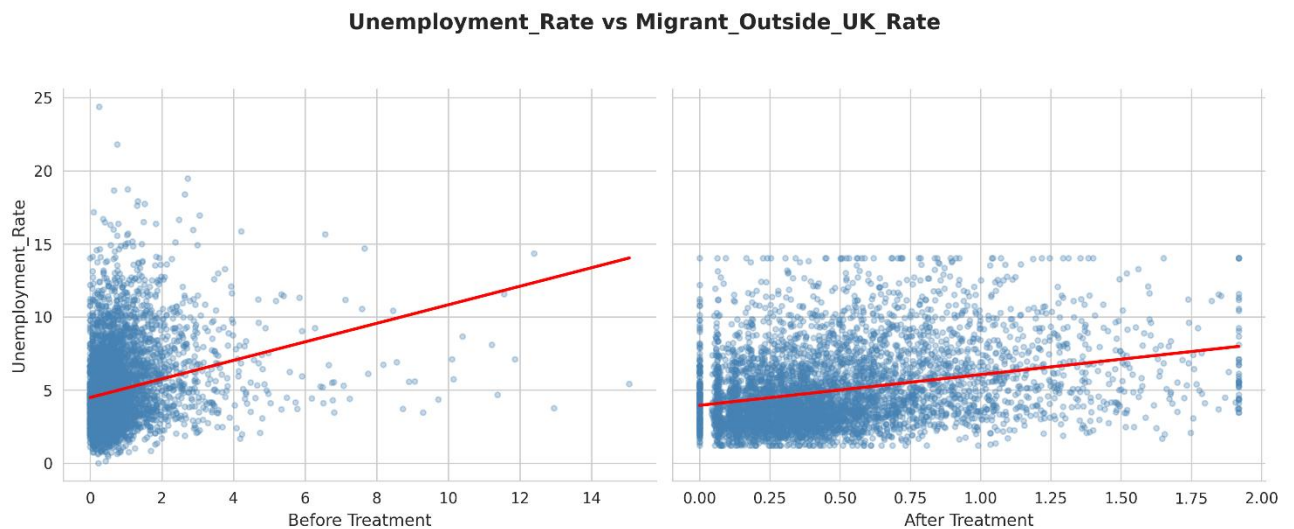


Figure 4.8: Relationship Between Unemployment Rate and Migrant Outside UK Rate Before and After Data Treatment

In comparison, the relationship between migrant outside UK rate and unemployment was less pronounced. The variable was nevertheless retained for multivariable assessment because migration represents a specific dimension of the research questions and its contribution needed to be assessed alongside deprivation and other socio-economic characteristics.

The remaining histograms, density plots, boxplots and predictor-versus-target plots generated during EDA are presented in the link provided in the Appendix to avoid overcrowding the Implementation chapter.

4.3.2. Correlation and Multicollinearity Assessment

Pearson correlation analysis was implemented to assess relationships between unemployment and candidate predictors and to identify highly overlapping variables.

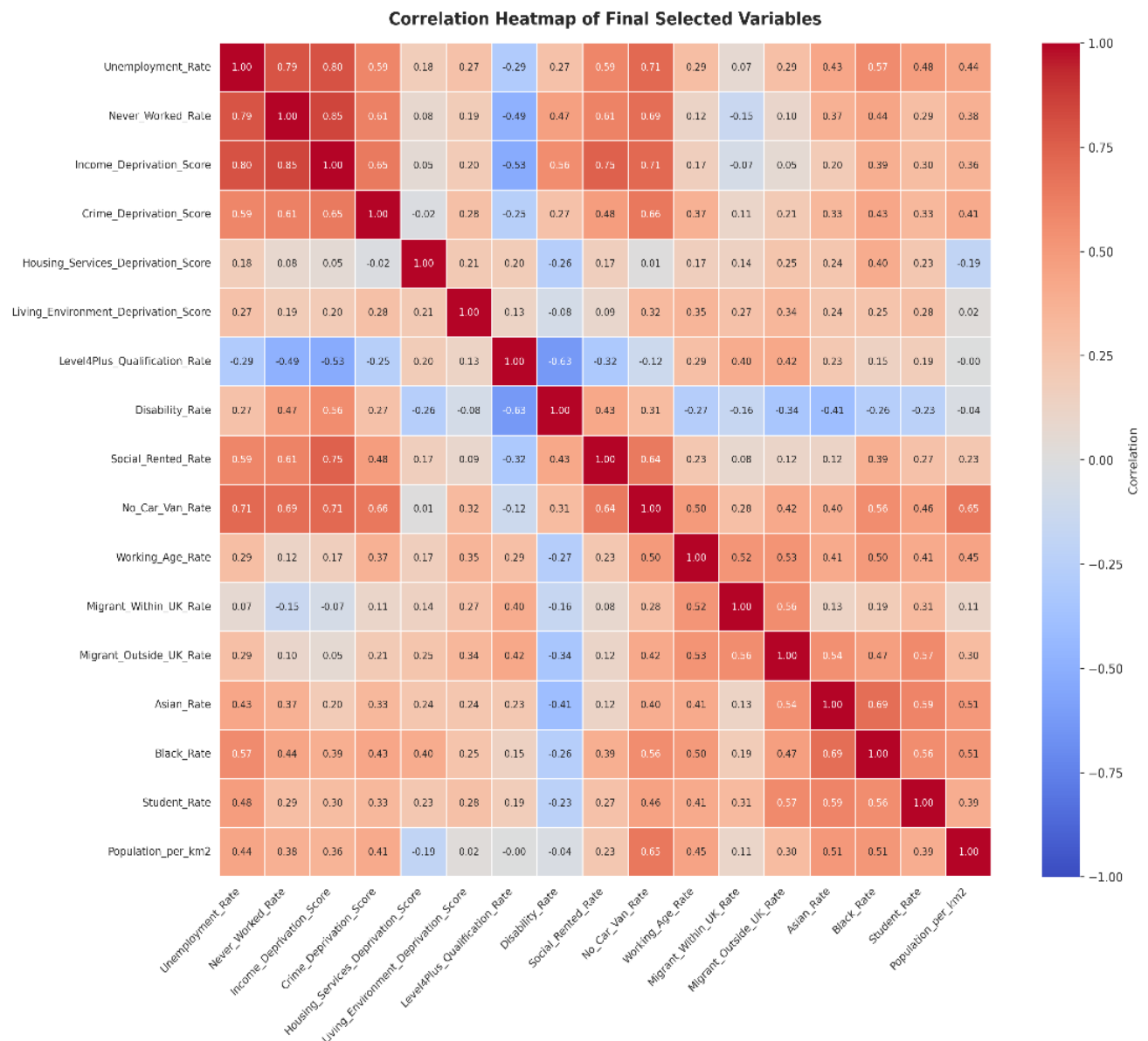


Figure 4.9: Correlation Heatmap of the Final Selected Variables

The heatmap showed strong relationships between unemployment and several socio-economic indicators. Income Deprivation Score and Never Worked Rate showed the strongest positive associations with unemployment, while No Car or Van Rate also demonstrated a strong positive relationship. In contrast, qualification-related indicators generally showed weaker or negative associations.

Correlation screening was followed by Variance Inflation Factor (VIF) analysis to assess multicollinearity among the candidate predictors. Variables representing duplicated or strongly overlapping concepts were reviewed before the final modelling dataset was established. The final sixteen predictors produced VIF values between approximately 1.51 and 7.26, with No Car or Van Rate and Income Deprivation Score producing the highest values. These results

were retained because the final selection balanced statistical screening with the socio-economic domains required by the research framework.

Domain	Final variables
Employment	Never Worked Rate
Deprivation	Income, Crime, Housing & Services, Living Environment
Education	Level 4+ Qualification Rate
Health	Disability Rate
Housing	Social Rented Rate
Transport	No Car or Van Rate
Demography	Working Age Rate, Population Density
Migration	Migrant Within UK Rate, Migrant Outside UK Rate
Ethnicity	Asian Rate, Black Rate
Economic activity	Student Rate

Table 4.3: Final Variables Selected for Machine Learning

The final predictor set therefore resulted from the combined literature-based review, correlation screening and multicollinearity assessment rather than automated statistical selection alone.

4.4. Machine Learning Implementation

The final modelling dataset was divided into 80% training data and 20% testing data using a fixed random state of 42. The same split was retained across all algorithms to ensure that performance comparisons were based on identical training and testing observations. Predictor standardisation was applied for Linear Regression, while the treated unscaled predictors were used for the tree-based models.

Linear Regression was implemented as the baseline model, followed by Random Forest, XGBoost and LightGBM. Default versions of the three ensemble models were first evaluated. Hyperparameter tuning was then implemented using GridSearchCV with three-fold cross-validation. The search examined model-specific parameters including the number of estimators, tree depth, learning rate and sampling controls. This produced default and tuned versions of Random Forest, XGBoost and LightGBM, allowing the effect of optimisation to be assessed rather than reporting only the best-performing configuration.

4.5. Model Evaluation and Interpretation

Model predictions were evaluated using RMSE, MAE, R^2 and Adjusted R^2 .

Model	RMSE	MAE	R ²	Adjusted R ²
XGBoost Tuned	1.1033	0.8391	0.8207	0.8202
XGBoost	1.1047	0.8439	0.8202	0.8198
LightGBM	1.1087	0.8423	0.8189	0.8185
LightGBM Tuned	1.1090	0.8438	0.8188	0.8184
Random Forest Tuned	1.1190	0.8495	0.8155	0.8151
Random Forest	1.1247	0.8532	0.8137	0.8132
Linear Regression	1.2626	0.9524	0.7652	0.7646

Table 4.4: Model Performance Comparison

The implementation identified tuned XGBoost as the strongest-performing configuration, producing the lowest RMSE and MAE and the highest R². The complete performance values are recorded in the current Chapter 4.

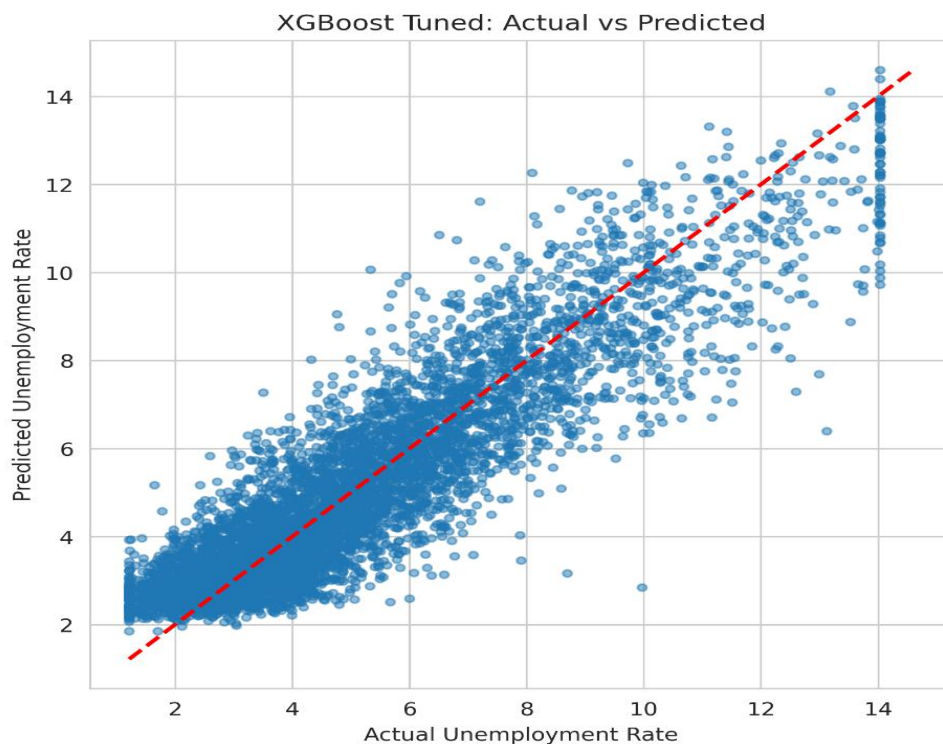


Figure 4.10: Actual vs Predicted Unemployment Rate for Tuned XGBoost

The concentration of observations around the reference line provides visual evidence of agreement between observed and predicted unemployment values.

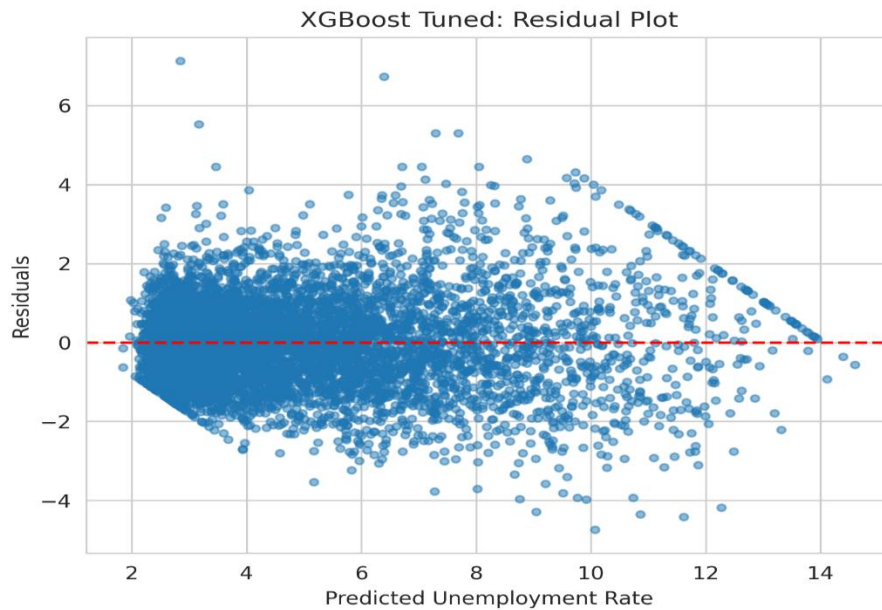


Figure 4.11: Residual Plot for Tuned XGBoost

The residual plot shows prediction errors distributed predominantly around zero, while larger residuals remained for some observations.

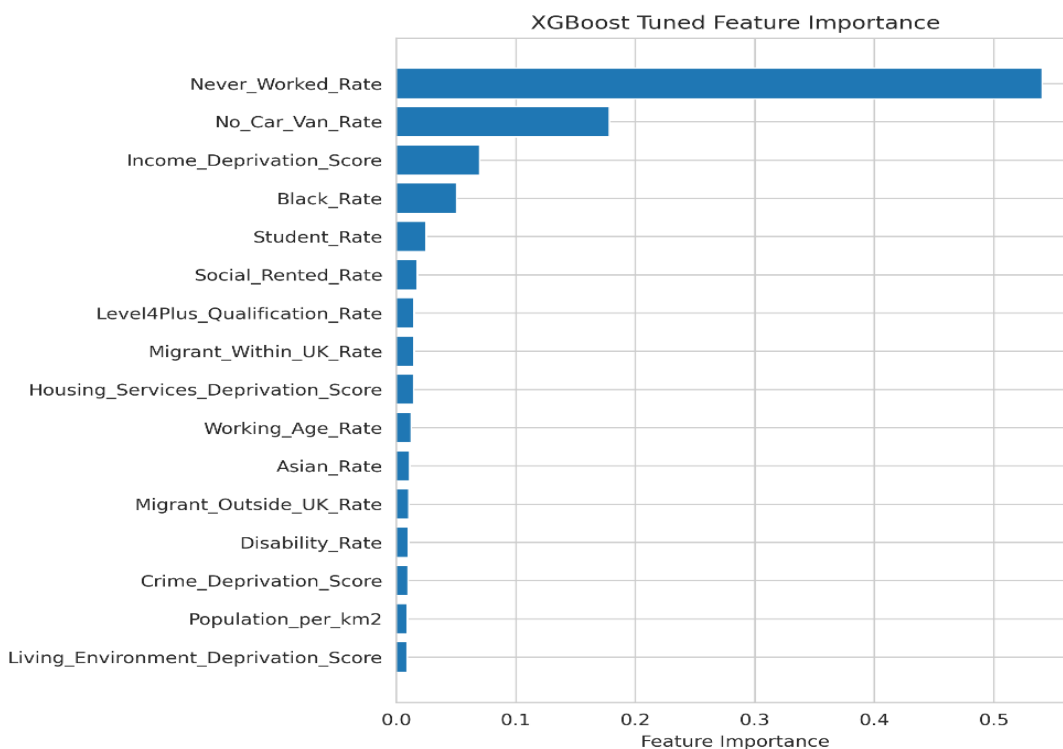


Figure 4.12: Feature Importance for Tuned XGBoost

Never Worked Rate was the dominant XGBoost feature, while No Car or Van Rate and Income Deprivation Score also ranked highly. SHAP analysis was subsequently applied to provide greater transparency concerning the contribution and direction of individual predictors.

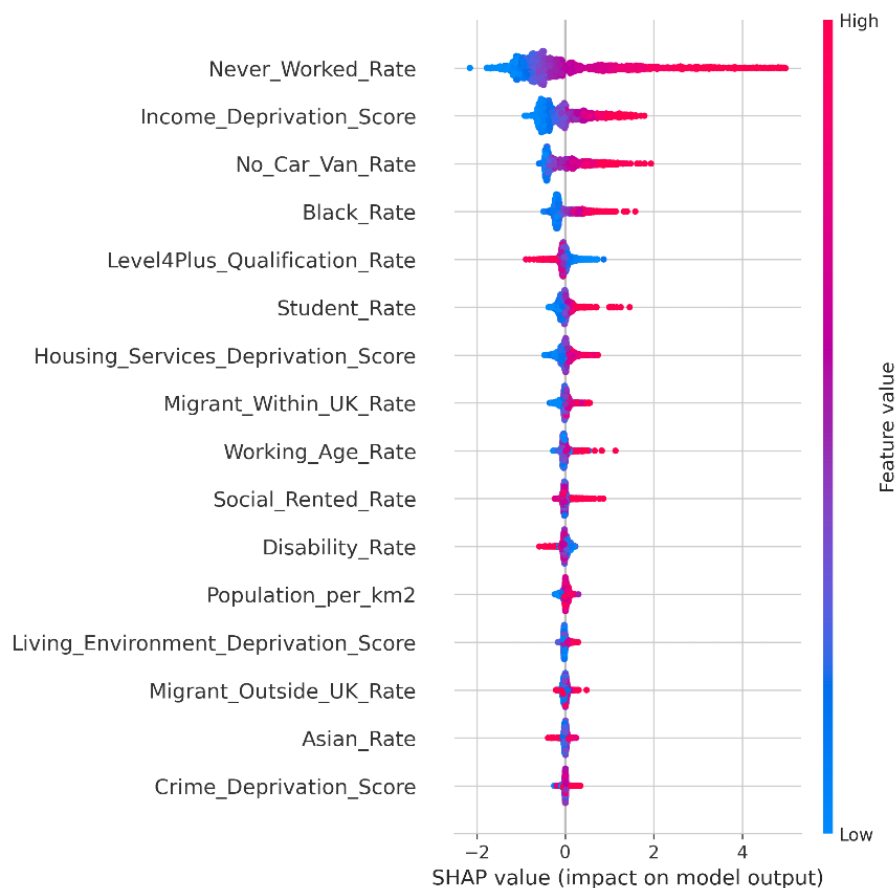


Figure 4.13: SHAP Summary Plot

The SHAP output reinforced the importance of Never Worked Rate, Income Deprivation Score and No Car or Van Rate and demonstrated variation in how predictor values contributed to model predictions. The current implementation already includes feature-importance and SHAP outputs as part of the interpretation stage.

4.6. Spatial Analysis Implementation

The final analytical data were exported to QGIS and joined with official England LSOA boundaries using LSOA_Code. Individual choropleth maps were produced for unemployment and selected high-ranking socio-economic predictors.

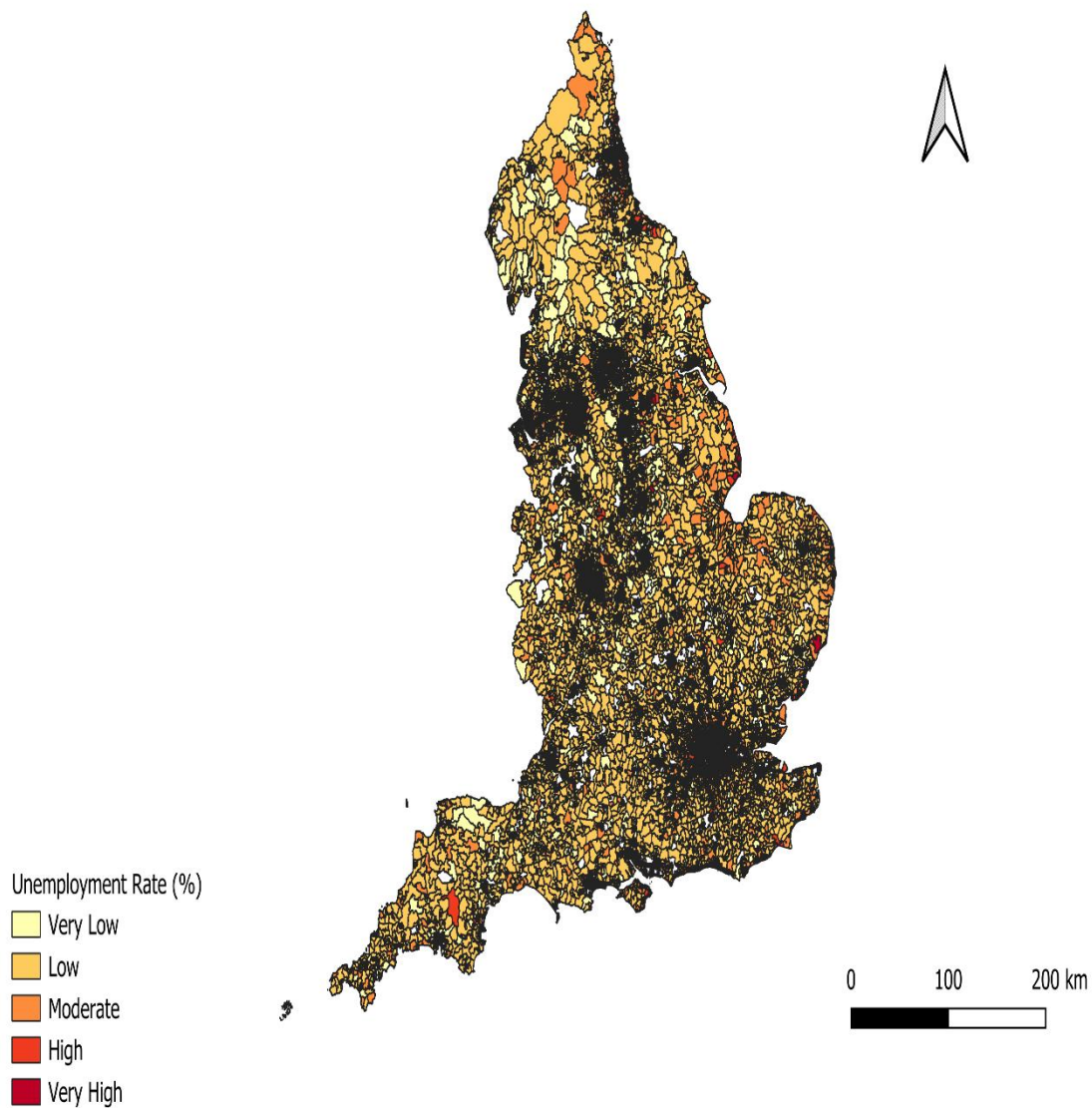


Figure 4.14: Spatial Distribution of Unemployment Rate

The above figure presents the spatial distribution of unemployment across English LSOAs. The map ranges from very low in light yellow to very high in dark red, demonstrating substantial geographical variation in unemployment. Higher unemployment appears concentrated in particular urban and coastal LSOAs, while many less densely populated areas show lower rates. This indicates that unemployment is spatially uneven rather than uniformly distributed across England.

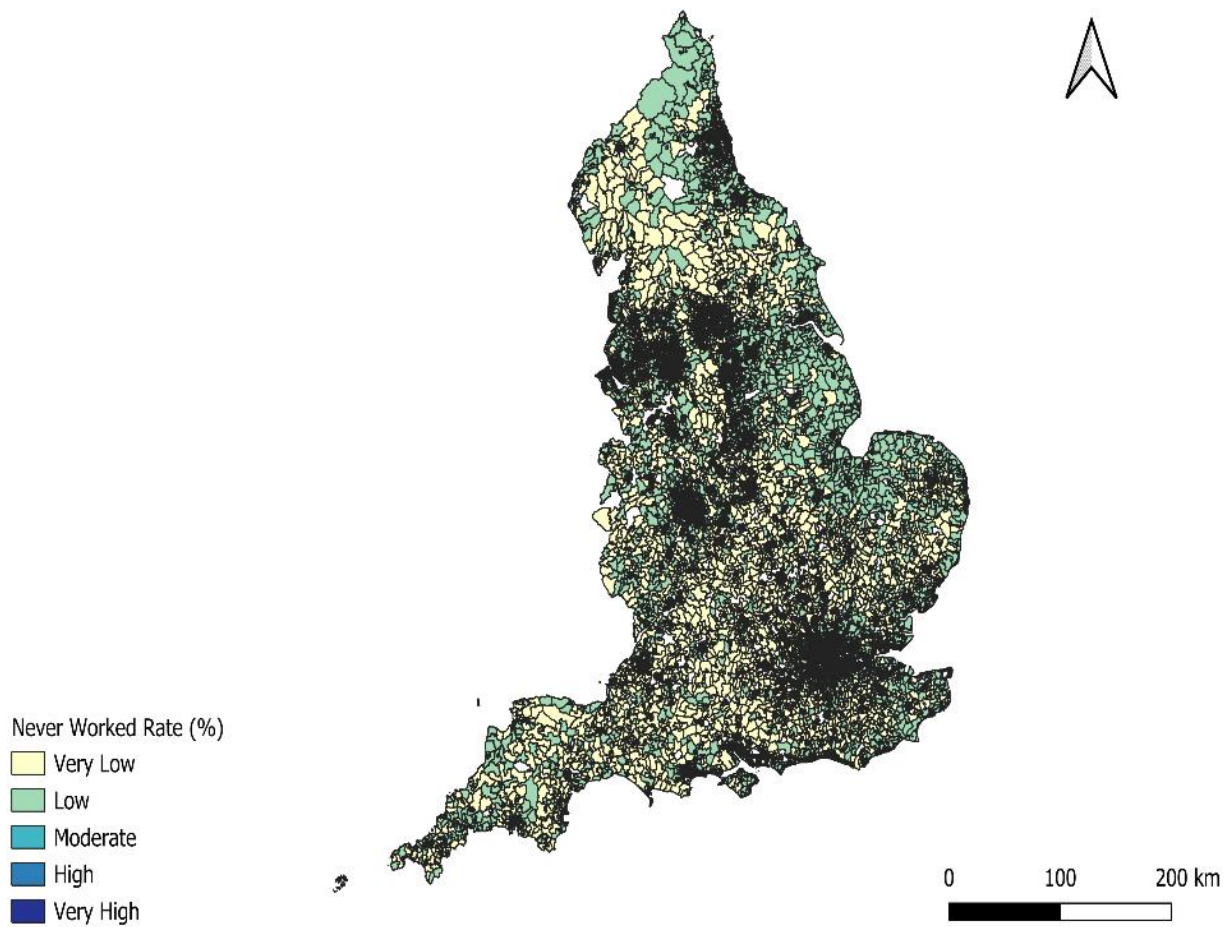


Figure 4.15: Spatial Distribution of Never Worked Rate

The above figure shows the spatial distribution of the Never Worked Rate. Light yellow and green represent lower rates, while darker blue represents higher rates. Higher values appear more concentrated in particular urban LSOAs, whereas lower values are more widely visible across less urbanised areas. The geographical variation suggests that long-term labour-market detachment is concentrated in particular neighbourhoods rather than being evenly distributed.

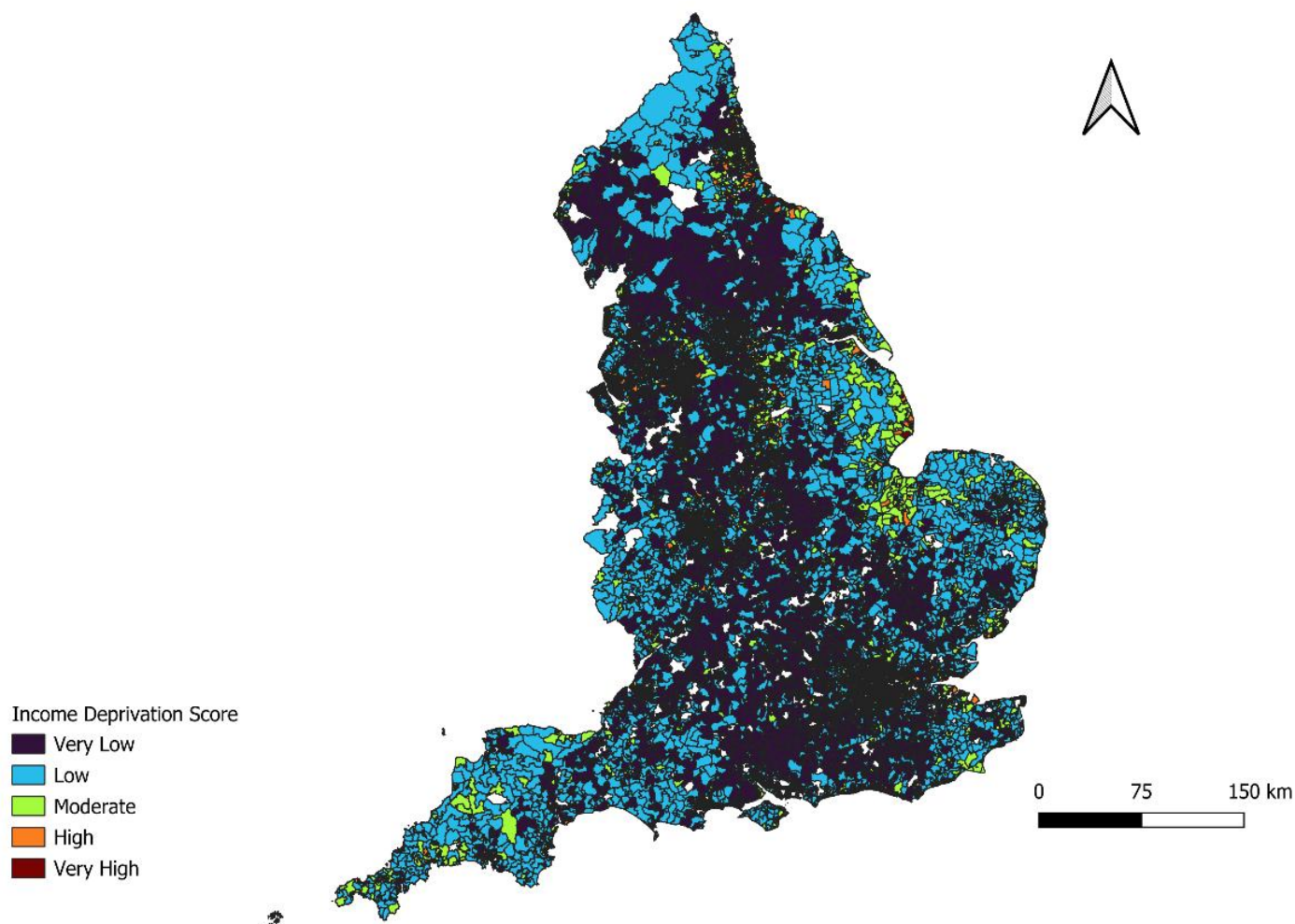


Figure 4.16: Spatial Distribution of Income Deprivation Score

The above figure presents the spatial distribution of income deprivation across England. Dark purple represents very low deprivation, while green, orange and dark red indicate progressively higher deprivation. Higher deprivation occurs in identifiable pockets, particularly within and around some urban and coastal areas, while lower deprivation is widespread across other LSOAs. This spatial contrast demonstrates that economic disadvantage is geographically concentrated and may contribute to the observed variation in unemployment.

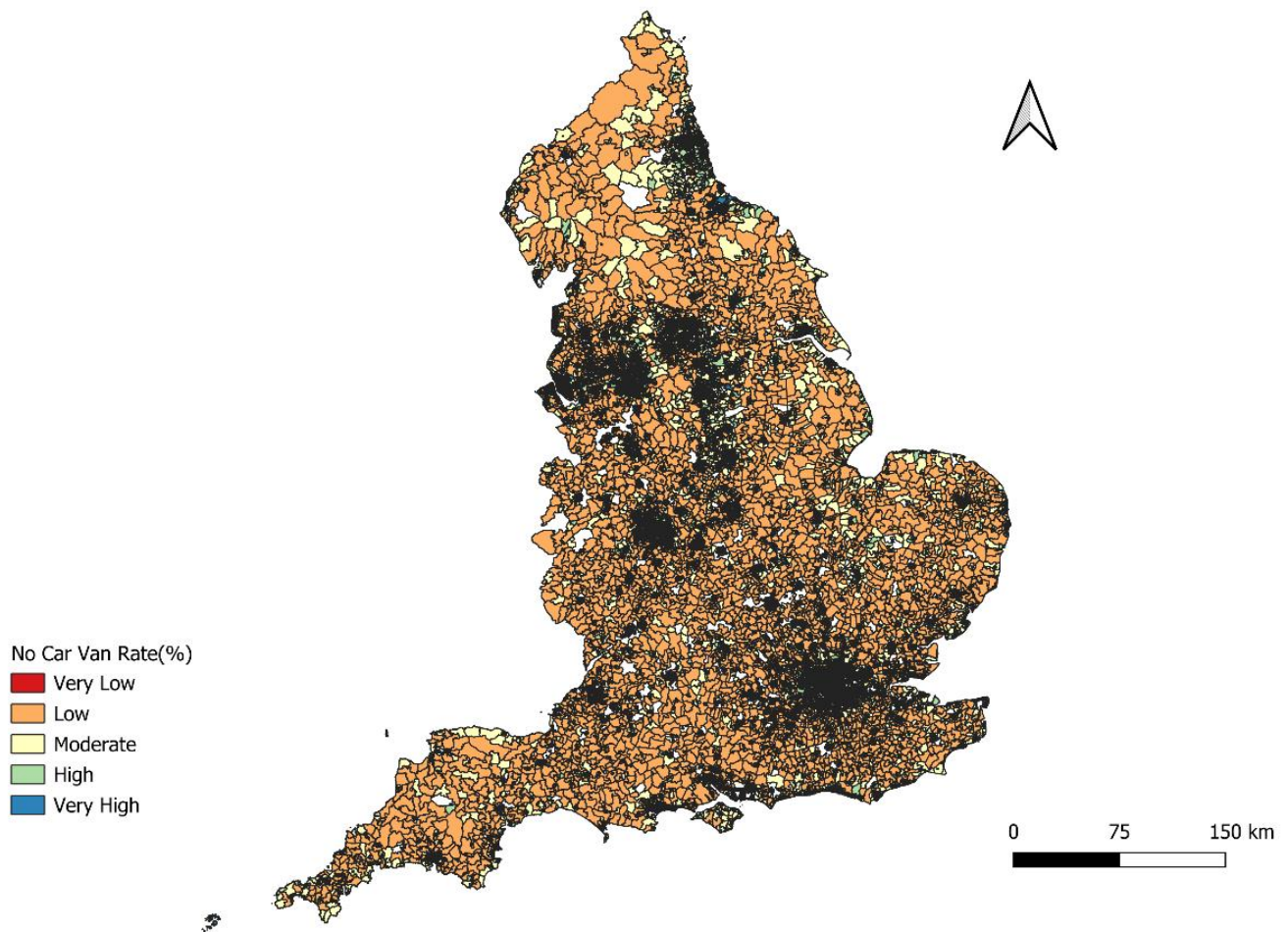


Figure 4.17: Spatial Distribution of No-Car Van Rate

The above figure shows geographical variation in the proportion of households with no car or van. The classification ranges from very low in red through orange and yellow to high and very high in green and blue. Higher rates are visible in particular densely populated urban areas, while lower rates are more widespread outside these concentrations. This pattern suggests a geographical relationship between household transport availability and wider socio-economic conditions.

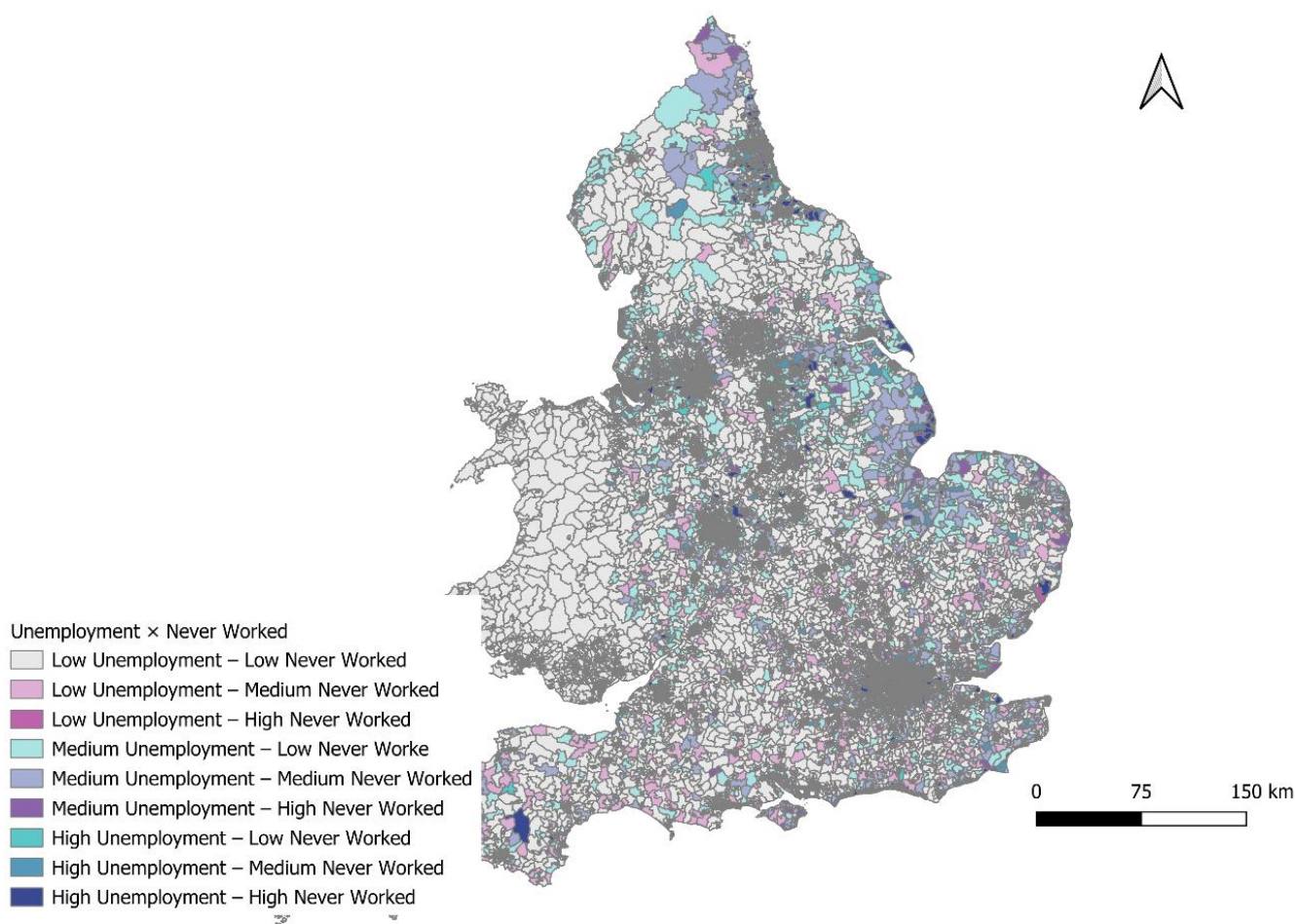


Figure 4.18: Bivariate Map; Unemployment Rate and Never Worked Rate

The above figure combines unemployment and Never Worked Rate using nine categories. Light grey represents low unemployment and low Never Worked Rate, while dark blue represents high unemployment and high Never Worked Rate. Darker blue LSOAs occur in identifiable geographical clusters, whereas light-grey areas indicate neighbourhoods where both measures are comparatively low. The co-location of high values supports the strong positive relationship identified in the statistical analysis and suggests that labour-market detachment and unemployment share a clear spatial dimension.



Figure 4.19: Bivariate Map: Unemployment Rate and Income Deprivation Score

The above figure combines unemployment with income deprivation. Light grey represents low unemployment and low income deprivation, while the darkest green represents high unemployment and high income deprivation. Dark-green areas identify LSOAs where the two disadvantages coincide, while light-grey areas indicate comparatively favourable conditions for both measures. The presence of these high-high concentrations provides spatial evidence consistent with the strong positive correlation between income deprivation and unemployment.

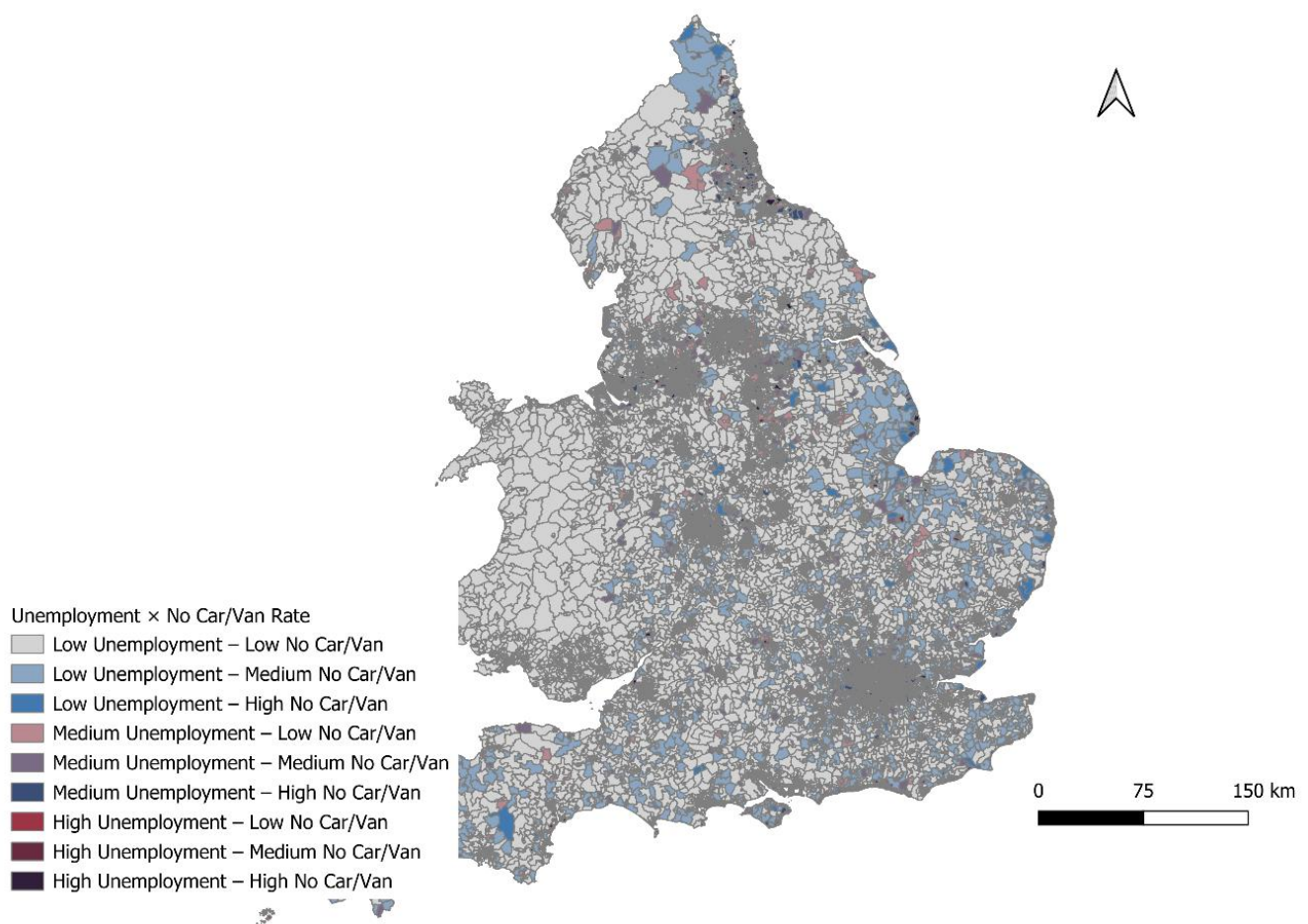


Figure 4.20: Bivariate Map: Unemployment Rate and No Car or Van Rate

The above figure combines unemployment with the No Car or Van Rate. Light grey represents low unemployment and low No Car/Van Rate, while the darkest purple represents high unemployment and high No Car/Van Rate. Dark-purple LSOAs identify areas experiencing both high unemployment and relatively limited household car availability, while light-grey areas have low levels of both. Their spatial overlap supports the statistical association between unemployment and lack of car access, although this relationship should be interpreted as association rather than direct causation.

4.7. Chapter Summary

This chapter presented the implementation of the analytical framework developed in Chapter 3. Fifteen Census datasets covering 35,672 LSOAs in England and Wales were combined with England-specific socio-economic data to produce a final analytical dataset of 31,810 English LSOAs. Data quality inspection, feature engineering, data treatment, exploratory data analysis, correlation analysis and VIF analysis were conducted to prepare a final set of sixteen predictors for modelling.

Linear Regression, Random Forest, XGBoost and LightGBM were implemented within a common evaluation framework, with tuned XGBoost achieving the strongest predictive performance. Feature importance and SHAP were used to identify the variables contributing to model predictions, while QGIS was used to visualise selected relationships at the neighbourhood level. These results form the empirical basis for the critical interpretation and discussion in Chapter 5.

5. CHAPTER 5: DISCUSSION AND ANALYSIS

5.1. Introduction

Chapter 4 presented the empirical outputs from the statistical, machine-learning and spatial analyses. This chapter critically interprets these findings in relation to the research aim, research questions and literature reviewed in Chapter 2. The discussion focuses on socio-economic relationships with unemployment, the relative contribution of migration, comparative model performance, predictor influence and geographical patterns across English Lower Layer Super Output Areas (LSOAs).

The following sections address each research question in turn, linking the empirical evidence from Chapter 4 with previous research before informing the conclusions and recommendations presented in Chapter 6.

5.2. Socio-economic Characteristics and Unemployment

RQ1: Which socio-economic characteristics are most strongly associated with unemployment across English LSOAs?

The findings indicate that neighbourhood unemployment is closely associated with wider socio-economic disadvantage rather than with a single independent characteristic. Among the final predictors, Income Deprivation Score, Never Worked Rate and No Car or Van Rate showed particularly strong positive relationships with unemployment, while qualification-related characteristics generally showed weaker or negative relationships. This pattern suggests that differences in unemployment across English LSOAs reflect interconnected dimensions of labour-market participation, deprivation, accessibility and human capital.

The strong relationship with income deprivation is consistent with Lloyd, Norman and McLennan (2023), who identified persistent geographical concentrations of small-area disadvantage in England, and with Knies and Kumari (2022), who demonstrated that individual deprivation domains contain distinct information. However, the present study extends these findings by comparing several deprivation dimensions simultaneously within an unemployment prediction framework. Income Deprivation Score was more influential than several other deprivation measures, suggesting that different dimensions of deprivation are not equally informative for neighbourhood unemployment. This provides support for examining domain-specific measures rather than relying exclusively on a composite deprivation index.

Nevertheless, the strength of this relationship should not be interpreted causally. Income deprivation and unemployment are likely to be mutually connected: unemployment can reduce household resources, while neighbourhoods experiencing persistent economic disadvantage may also have weaker employment opportunities. Income deprivation may additionally capture conditions reflected through housing, accessibility and other socio-economic characteristics. Its predictive importance therefore indicates a strong structural association but does not establish income deprivation as an independent cause of unemployment.

Never Worked Rate showed an even stronger association with persistent labour-market disadvantage. Its positive relationship with unemployment remained after data treatment and it became highly influential within XGBoost. This indicates that neighbourhoods with higher concentrations of people who have never worked also tend to experience higher current

unemployment. However, greater caution is needed when interpreting this result, as Never Worked Rate and unemployment both reflect related aspects of labour-market disengagement. The high model importance of Never Worked Rate may therefore partly reflect its conceptual proximity to the dependent variable, rather than a distinct explanatory factor. This finding is best interpreted as evidence of the geographical persistence and concentration of labour-market exclusion.

The prominence of No Car or Van Rate introduces an accessibility dimension and is broadly consistent with Bastiaanssen, Johnson and Lucas (2022), who found that transport accessibility was associated with employment, particularly among lower-income and low-car-ownership groups. However, the comparison is not direct, as their study assessed employment using individual-level accessibility measures, while the present research uses neighbourhood-level vehicle availability. High No Car or Van Rate may indicate mobility constraints, but it may also serve as a proxy for income, urban context or wider deprivation. In well-connected urban neighbourhoods, low vehicle ownership may have limited impact on employment, while in less accessible areas it may restrict access to more dispersed opportunities.

Education showed a contrasting relationship. Qualification-related measures generally displayed weaker or negative associations with unemployment, which is broadly consistent with Overman and Xu (2024), who emphasise skills as an important component of geographical labour-market inequality. Higher qualifications may increase access to a wider range of occupations and provide greater flexibility within changing labour markets. However, the weaker relative importance of education in the present models suggests that human capital alone does not fully account for neighbourhood unemployment where structural disadvantage and limited employment accessibility remain significant.

RQ1 is therefore answered by identifying labour-market detachment, income deprivation and limited car availability as the strongest socio-economic signals associated with unemployment, while educational advantage appears comparatively protective. Importantly, comparison with previous research suggests that these findings are broadly consistent with established evidence, while the LSOA-level analysis demonstrates their relative importance when evaluated together. The results therefore reinforce the view that neighbourhood unemployment reflects overlapping socio-economic conditions rather than a single isolated determinant.

5.3. Migration in the Wider Socio-economic Context

RQ2: What predictive contribution do migration indicators make when considered alongside other socio-economic characteristics?

Migration was retained in the analytical framework because RQ2 aimed to assess whether migration remained predictively informative when considered alongside wider structural socio-economic conditions, rather than whether it had the strongest simple relationship with unemployment. The results in Chapter 4 showed that the relationship between Migrant Outside UK Rate and unemployment was less pronounced than for Never Worked Rate, indicating that migration represents only one aspect of the broader neighbourhood labour-market context.

Both Migrant Within UK Rate and Migrant Outside UK Rate contributed information to the final modelling framework, but neither was among the dominant predictors. Labour-market detachment, income deprivation and limited car availability provided substantially stronger predictive signals. This suggests that differences in neighbourhood unemployment are more

strongly associated with structural socio-economic conditions than with migration composition alone. However, lower predictive importance does not indicate that migration has no relationship with labour-market outcomes.

The findings can be compared with Clark et al. (2019), who found that employment outcomes among migrants to England varied across migrant groups and local contexts. Jivraj and Alao (2023) similarly identified persistent labour-market inequalities among some ethnic groups but demonstrated that these patterns could not be explained solely by neighbourhood deprivation or ethnic density. The present findings broadly support the contextual interpretation offered by these studies because migration does not emerge as an isolated or dominant explanation of unemployment. However, direct comparison requires caution. These previous studies examined individual employment outcomes and differences between population groups, whereas this research examines unemployment rates and socio-economic characteristics aggregated at LSOA level. The lower importance of migration in the present models therefore does not contradict evidence of individual labour-market inequalities.

The relatively modest contribution of the migration variables may also reflect limitations in how migration is represented. LSOA-level rates do not distinguish migrants by qualifications, occupation, age, duration of residence, reason for migration or country of origin. These characteristics may substantially influence employment outcomes, so aggregation can conceal heterogeneity between migrant populations. More detailed individual-level or longitudinal data could therefore produce different conclusions about the relationship between migration and employment.

An additional ecological limitation must be recognised. An association between the migrant composition of an LSOA and its unemployment rate does not demonstrate that migrants within that neighbourhood are themselves unemployed. Such an interpretation would incorrectly transfer an area-level relationship to individuals and would exceed what the data can support.

RQ2 therefore leads to a more qualified conclusion. Migration contributes predictive information, but its contribution is secondary to the stronger structural characteristics identified in this research. Compared with the literature, these findings reinforce the importance of examining migration within its wider socio-economic and geographical context rather than treating migrant concentration as an independent explanation of neighbourhood unemployment.

5.4. Comparative Predictive Performance

RQ3: Which modelling approach provides the strongest predictive performance for unemployment across English LSOAs?

The model comparison identified tuned XGBoost as the strongest-performing approach, with an R^2 of 0.8207. However, its advantage over the other ensemble models was relatively small. Default XGBoost achieved an R^2 of 0.8202, default LightGBM 0.8189, tuned LightGBM 0.8188, tuned Random Forest 0.8155 and default Random Forest 0.8137, compared with 0.7652 for Linear Regression. The consistent advantage of all three ensemble approaches over the linear benchmark is therefore more significant than the marginal differences between individual tree-based models. This suggests that the ensemble methods represented variation in neighbourhood unemployment more effectively, without implying that XGBoost was substantially superior.

The performance difference between Linear Regression and the ensemble models also provides insight into the structure of the relationships identified in this research. A linear additive model assumes comparatively simple relationships between predictors and the outcome, whereas neighbourhood socio-economic characteristics may interact or have nonlinear effects. For example, limited car availability may have different implications depending on deprivation, population density or local accessibility. Tree-based models can represent such interactions and nonlinear patterns without requiring them to be specified in advance. Their stronger performance therefore suggests that neighbourhood unemployment is associated with relationships not fully captured by a simple linear structure.

This result can be compared with the machine-learning literature reviewed in Chapter 2. Gogas, Papadimitriou and Sofianos (2022) reported strong Random Forest performance in unemployment prediction, while Tu and Tran (2025) found XGBoost effective for UK unemployment forecasting. Celbiş (2023) similarly demonstrated the ability of tree-based machine learning to identify heterogeneous labour-market relationships. The findings of this research therefore broadly support previous evidence that nonlinear approaches can offer predictive advantages when labour-market outcomes depend on complex combinations of characteristics.

Nevertheless, agreement with previous studies should not be interpreted as evidence that XGBoost is universally preferable for unemployment modelling. Gogas, Papadimitriou and Sofianos (2022) examined an aggregate forecasting problem, while Tu and Tran (2025) used UK monthly time-series data. Celbiş (2023) also examined a different population and geographical context. In contrast, this research predicts cross-sectional differences across English LSOAs. Differences in geographical scale, predictor composition, outcome definition and modelling design mean that performance values cannot be directly compared across these studies. The evidence establishes only that XGBoost performed best within the dataset and evaluation framework adopted in this research.

The relatively small differences between XGBoost, LightGBM and Random Forest provide a further important finding. Hyperparameter tuning produced only a modest improvement for XGBoost and did not improve LightGBM. Greater model complexity or additional tuning did not automatically generate meaningful predictive gains. This suggests that the relevance and quality of the socio-economic information supplied to the models may be at least as important as small differences between ensemble algorithms.

The best-performing model did not explain all variation in unemployment, and the residual analysis identified larger errors for some observations. The remaining prediction error may reflect local labour-market conditions not represented by the selected Census and deprivation indicators, such as labour demand, vacancies, wage conditions, industrial restructuring and access to specific employment opportunities. Strong predictive performance should therefore not be interpreted as a complete representation of the processes shaping neighbourhood unemployment.

RQ3 is therefore answered by identifying tuned XGBoost as the strongest-performing model within this research. More importantly, comparison across all four modelling approaches indicates that nonlinear ensemble models provide a stronger representation of neighbourhood unemployment than the linear benchmark. The broader contribution is evidence of nonlinear predictive structure rather than simply the identification of a single winning algorithm.

5.5. Most Influential Predictors

RQ4: Which variables have the greatest influence on predictions produced by the best-performing model?

Interpretation of the tuned XGBoost model identifies Never Worked Rate as the most influential predictor, followed by No Car or Van Rate and Income Deprivation Score. SHAP analysis supports the importance of these variables and shows that their contributions differ across individual LSOAs. This suggests that the model does not apply the influence of each socio-economic characteristic equally across all neighbourhoods.

The consistency across analytical methods strengthens this interpretation. Correlation analysis identified Income Deprivation Score, Never Worked Rate and No Car or Van Rate as strongly related to unemployment. XGBoost feature importance also ranked these variables highly, and SHAP confirmed their contribution to individual model predictions. This convergence provides stronger evidence of their predictive relevance than reliance on a single importance measure. It also reflects the literature reviewed in Chapter 2, where deprivation, labour-market disadvantage and accessibility were identified as important dimensions of labour-market inequality.

However, agreement between correlation, feature importance and SHAP does not establish causal importance. As Lundberg and Lee (2017) demonstrate, SHAP explains the contribution of features to predictions generated by a fitted model rather than the underlying causal processes. In addition, highly ranked variables may reflect overlapping socio-economic conditions. Never Worked Rate is conceptually close to unemployment, while No Car or Van Rate may capture both accessibility constraints and broader economic disadvantage. Their importance should not be interpreted as evidence that directly changing these characteristics would necessarily reduce unemployment.

The findings instead indicate that the tuned XGBoost model consistently relies on information representing labour-market attachment, economic disadvantage and accessibility. The alignment with themes identified in the literature strengthens their substantive relevance, although differences in measurement and analytical scale require cautious interpretation.

RQ4 therefore identifies theoretically meaningful predictors within the strongest-performing model, rather than treating XGBoost as an uninterpretable predictive system. However, their influence remains model-specific, predictive and associative rather than causal, which limits their direct interpretation as independent policy intervention targets.

5.6. Geographical Patterns of Unemployment and Disadvantage

RQ5: What geographical patterns of unemployment and its key socio-economic correlates can be identified using QGIS?

The QGIS results demonstrate that the relationships identified statistically also have a geographical dimension. Unemployment and its leading socio-economic correlates are unevenly distributed across English LSOAs, supporting the argument developed in Chapter 2 that labour-market inequality is spatially differentiated and reinforcing the value of neighbourhood-level analysis.

The bivariate analysis of Unemployment Rate and Never Worked Rate reveals clear high-high and low-low patterns. The concentration of LSOAs with both high unemployment and high Never Worked Rate suggests that persistent labour-market detachment is closely associated with current unemployment. However, as these indicators reflect related aspects of labour-market disadvantage, interpretation should remain cautious.

Income deprivation shows a similar geographical relationship. Neighbourhoods with both high unemployment and high Income Deprivation Score reinforce the strong statistical association identified in the correlation and modelling analysis. This aligns with Lloyd, Norman and McLennan (2023), who found persistent geographical concentrations of small-area disadvantage in England. The present research extends this evidence by demonstrating their co-occurrence at LSOA level.

The relationship with No Car or Van Rate is more context-dependent. In well-connected urban areas, low vehicle ownership may indicate access to public transport, while in less accessible neighbourhoods it may represent a greater constraint on employment opportunities. The spatial pattern therefore supports an accessibility interpretation, but should not be interpreted as a direct effect of vehicle ownership on unemployment.

These findings broadly correspond with Gilmartin and Korobilis (2012), who identified geographical differences in unemployment, and Radburn and Beecham (2021), who showed that fine-scale mapping can reveal localised disadvantage that broader geographical analysis may obscure. However, direct comparison is limited by differences in geographical units, variables and analytical approaches.

An important limitation is that the QGIS maps show geographical co-occurrence rather than statistically verified spatial dependence. Visible high-high patterns do not establish spatial clustering or spillover effects. Formal spatial analysis, such as Global Moran's I and Local Indicators of Spatial Association, would be required to test whether these concentrations are statistically significant.

RQ5 therefore shows that unemployment and its principal socio-economic correlates are geographically uneven across England. The LSOA-level analysis complements the predictive modelling by identifying where important socio-economic conditions coincide, while the observed patterns should be interpreted as spatial co-occurrence rather than formally established spatial dependence.

5.7. Overall Interpretation

Across the five research questions, unemployment in English LSOAs is most strongly associated with labour-market detachment, income deprivation and limited car availability, while migration contributes less when broader socio-economic conditions are included. These findings support the literature reviewed in Chapter 2 by demonstrating that neighbourhood unemployment is a multidimensional outcome rather than one explained by a single characteristic.

The stronger performance of ensemble models suggests that these relationships include nonlinearities and interactions not fully captured by Linear Regression. However, the small differences between the ensemble models indicate that the general advantage of nonlinear modelling is more significant than XGBoost's marginally stronger performance. QGIS analysis

further shows that the identified disadvantages are geographically uneven, although the maps represent geographical co-occurrence rather than statistically established spatial dependence.

This research addresses the identified gap by combining multiple socio-economic dimensions, comparative predictive modelling and spatial analysis within a consistent England-wide LSOA framework. However, differences in geographical scale, data and methodology mean that caution is required when comparing these findings directly with previous research.

From a practical perspective, the models may support neighbourhood-level decision making but are not yet suitable for direct operational deployment. More frequently updated data on vacancies, claimant counts and local labour demand, along with temporal and geographical validation, would be required to improve practical reliability.

Overall, this research indicates that neighbourhood unemployment is multidimensional, predictively nonlinear and geographically uneven. However, the cross-sectional and aggregated design means that the findings represent predictive associations and spatial patterns rather than causal effects.

5.8. Chapter Summary

This chapter critically interpreted the findings presented in Chapter 4 in relation to the research questions and literature. The results identified labour-market detachment, income deprivation and limited car availability as the strongest predictors of neighbourhood unemployment, while migration made a smaller contribution when considered alongside wider socio-economic conditions. Tuned XGBoost achieved the strongest performance, although the similar results across ensemble models suggest that their broader advantage lies in capturing nonlinear relationships rather than the superiority of a single algorithm.

Feature importance and SHAP supported the relevance of the leading predictors, while QGIS demonstrated their geographical co-occurrence with unemployment. Overall, the five research questions indicate that unemployment across English LSOAs reflects interconnected socio-economic and geographical inequalities. Chapter 6 consolidates these findings and considers the research contributions, limitations and recommendations.

6. CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1. Conclusion

This study investigated the socio-economic characteristics associated with unemployment across English Lower Layer Super Output Areas (LSOAs) using machine learning and spatial analysis. Census 2021, IMD 2019 and population-density data were integrated to examine multiple dimensions of neighbourhood conditions within a consistent England-wide analytical framework.

Never Worked Rate, Income Deprivation Score and No Car or Van Rate emerged as the most influential characteristics associated with neighbourhood unemployment. Never Worked Rate was the dominant predictor in the best-performing model, but its conceptual proximity to unemployment suggests it is better interpreted as an indicator of persistent labour-market disengagement rather than as an independent determinant. Similarly, No Car or Van Rate may reflect accessibility, income and urban context rather than a direct effect of vehicle ownership. Migration indicators contributed predictive information but were less influential than the leading structural characteristics when considered alongside deprivation, education, housing, transport and other neighbourhood conditions. As these measures are aggregated at LSOA level, the findings do not establish the employment outcomes of individual migrants.

Tuned XGBoost achieved the strongest predictive performance, with R^2 of 0.8207, RMSE of 1.1033 and MAE of 0.8391, although its advantage over Random Forest and LightGBM was small. The ensemble models consistently outperformed Linear Regression, indicating that neighbourhood unemployment involves nonlinear relationships and interactions not fully captured by a simple linear structure.

QGIS analysis demonstrated substantial geographical variation in unemployment and its leading predictors. Bivariate maps identified neighbourhoods where high unemployment coincided with high Never Worked Rate, Income Deprivation Score or No Car or Van Rate, although these patterns represent spatial co-occurrence rather than statistically verified spatial dependence. Overall, the findings indicate that unemployment across English LSOAs is multidimensional, predictively nonlinear and geographically uneven. The principal contribution of this study is the integration of neighbourhood-level socio-economic analysis, comparative predictive modelling, model interpretation and spatial visualisation within a consistent England-wide framework.

6.2. Strengths and Limitations

The principal strengths of the research are:

- The England-wide LSOA design captures neighbourhood variation that broader regional analysis may conceal.
- Multiple socio-economic domains were evaluated within the same dataset, allowing their relative predictive contributions to be compared.
- Four models were evaluated under consistent conditions, providing a meaningful linear-versus-nonlinear comparison.

- Feature importance and SHAP improved interpretation rather than relying solely on predictive accuracy.
- QGIS complemented the statistical evidence by showing where unemployment and important socio-economic characteristics were geographically concentrated.

These strengths build on the integrated analytical framework described in the current dissertation.

The principal limitations are:

- The cross-sectional observational design supports association and prediction, not causal inference.
- Census 2021 and IMD 2019 have different reference periods, creating some temporal inconsistency.
- LSOA-level findings cannot be assumed to describe individual residents, particularly when interpreting migration and ethnicity.
- Never Worked Rate overlaps conceptually with unemployment, requiring caution when interpreting its high predictive importance.
- QGIS visualisation identifies geographical patterns but does not formally test spatial autocorrelation or spillover effects.
- The best-performing tuned XGBoost model achieved an RMSE of 1.1033 and R^2 of 0.8207, indicating strong predictive performance but also meaningful remaining prediction error, suggesting that relevant local labour-market factors may not be fully captured by the selected variables.
- The random 80/20 split demonstrates performance for unseen LSOAs within the dataset but does not establish generalisability to future periods or geographically distinct samples.
- Nearby LSOAs may share characteristics; therefore, future evaluation could apply spatial cross-validation or geographically blocked validation.
- The dataset lacks dynamic labour-market information such as vacancies, claimant counts and business activity, which may contribute to unexplained variation.

These limitations restrict the strength of interpretation but do not remove the value of the observed predictive and geographical patterns.

6.3. Recommendations

These recommendations are informed by the findings. First, neighbourhood employment support may be most effective when prioritised in areas where high unemployment coincides with persistent labour-market disengagement. The prominence of Never Worked Rate indicates potential value in targeted employability, skills and labour-market-entry support. However, this recommendation is based on observed associations rather than evidence that a specific intervention would directly reduce unemployment.

Second, the importance of Income Deprivation Score suggests that unemployment should be considered alongside wider neighbourhood disadvantage. Employment programmes may be more effective if coordinated with deprivation-focused support in areas where spatial evidence identifies overlapping disadvantage.

Third, the importance of No Car or Van Rate indicates that accessibility should be considered when targeting employment support. Local context is important, as low vehicle ownership may have different implications in well-connected urban areas compared to less accessible neighbourhoods.

The smaller predictive contribution of migration indicators suggests that migration levels alone are insufficient to explain or target neighbourhood unemployment. Greater attention should be given to wider socio-economic conditions, while recognising that aggregated data cannot identify barriers experienced by individual migrant groups.

Future research could incorporate more frequently updated labour-market information, including claimant counts, vacancies, earnings and local employment opportunities. Temporal and spatial validation may help assess whether model performance generalises across different periods and geographical areas. Formal techniques such as Moran's I, Local Indicators of Spatial Association and spatial regression could be used to test the spatial patterns identified through QGIS. Before operational deployment, model drift, fairness and interpretability should also be evaluated.

Overall, the findings support a place-sensitive and multidimensional approach, where machine learning identifies important predictive characteristics and spatial analysis highlights where these conditions coincide.

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APPENDIX

Appendix A: Source Code

The complete source code and project files used in this research are available through the following OneDrive link. These include the Python analysis workflow in Python and plain-text formats, the SQL code and database files used for data integration, and the QGIS project file used for spatial analysis and map generation.

Source Code and Project Files: [Source Code and Project Files](#)

Appendix B: Dataset

The dataset used for this dissertation, including the integrated and processed LSOA-level data used for machine learning and spatial analysis, is available through the following OneDrive link:

Dataset: [Datasets](#)

Appendix C: Complete Project Outputs

The complete project outputs generated during the analysis, including additional figures, model outputs, CSV result files, feature importance results, data-treatment outputs and other supporting materials not presented in the main dissertation, are available through the following OneDrive link:

Complete Project Files and Outputs: [DS7010 ML Project](#)